

USER'S MANUAL

FOR

IOT SHALA

Manufactured By

LOGSUN SYSTEMS

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RASPDUINO DEVELOPMENT BOARD

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1. Introduction of Raspberry Pi



Raspberry Pi-4

Raspberry Pi 4 Model B is the latest product in the popular Raspberry Pi range of computers. It offers ground-breaking increases in processor speed, multimedia performance, memory, and connectivity compared to the prior-generation Raspberry Pi 3 Model B+, while retaining backwards compatibility and similar power consumption. For the end user, Raspberry Pi 4 Model B provides desktop performance comparable to entry-level x86 PC systems.

This product's key features include a high-performance 64-bit quad-core processor, dual-display support at resolutions up to 4K via a pair of micro-HDMI ports, hardware video decode at up to 4Kp60, up to 8GB of RAM, dual-band 2.4/5.0 GHz wireless LAN, Bluetooth 5.0, Gigabit Ethernet, USB 3.0, and PoE capability (via a separate PoE HAT add-on).

Specification

Processor:

Broadcom BCM2711, quad-core Cortex-A72 (ARM v8)
64-bit SoC @ 1.5GHz

Memory:

1GB, 2GB, 4GB or 8GB LPDDR4
(depending on model) with on-die ECC

Connectivity:

2.4 GHz and 5.0 GHz IEEE 802.11b/g/n/ac wireless
LAN, Bluetooth 5.0, BLE
Gigabit Ethernet
2 × USB 3.0 ports
2 × USB 2.0 ports.

GPIO:

Standard 40-pin GPIO header
(fully backwards-compatible with previous boards)
Video & sound: 2 × micro HDMI ports (up to 4Kp60 supported)
2-lane MIPI DSI display port
2-lane MIPI CSI camera port
4-pole stereo audio and composite video port

Multimedia:

H.265 (4Kp60 decode);
H.264 (1080p60 decode, 1080p30 encode);
OpenGL ES, 3.0 graphics

SD card support:

Micro SD card slot for loading operating system
and data storage

Input power:

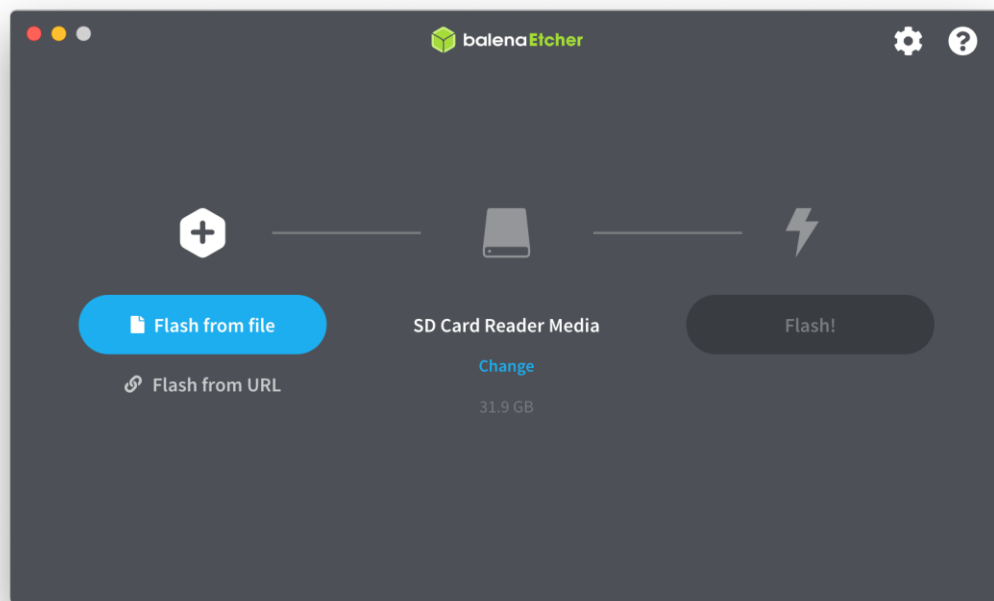
5V DC via USB-C connector (minimum 3A 1)
5V DC via GPIO header (minimum 3A1)
Power over Ethernet (PoE)–enabled
(requires separate PoE HAT)

Environment:

Operating temperature 0–50°C

2. OS INSTALLATION:

- 1 Download image (do not unzip) raspberry pi official website
<https://www.raspberrypi.com/software/operating-systems/>
- 2 Insert sd card to be flashed into computer
- 3 Download Balena Etcher Flashing Tool
- 3 Start belena Etcher



- 4 Click 'Flash from file'
- 5 Select your image file
- 6 Click 'Select target'
- 7 Select your sd card
- 8 Click 'Flash'

Booting Your Raspberry Pi for the First Time

After you're done writing the Raspberry Pi OS to a microSD card, it's time for the moment of truth.

1. **Insert the microSD card** into the Raspberry Pi.
2. **Connect the Raspberry Pi** to a monitor, keyboard and mouse.
3. **Connect an Ethernet cable** if you plan to use wired Internet.
4. **Plug the Pi in** to power it on.

If you had used the Raspberry Pi Imager settings to create a username and password, you'll be able to go straight into the desktop environment, but if not, you will get a setup wizard.

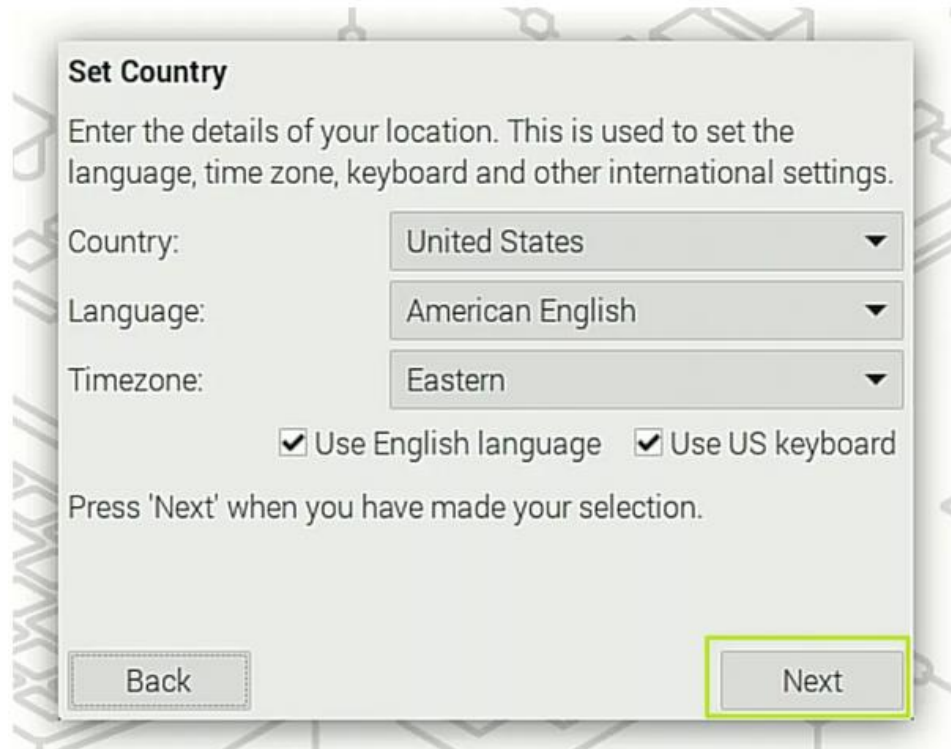
Using the Raspberry Pi First-Time Setup Wizard

If you chose a username and password in Raspberry Pi Imager settings, before writing the microSD card, you will get the desktop on first boot. But, if you did not, you'll be prompted to create a username and password and enter all the network credentials by a setup wizard on first boot. If that happens, follow these steps to finish setting up your Raspberry Pi.

1. **Click Next** on the dialog box.



2. **Set your country and language** and click Next. The default choices may already be the correct ones.



Set Country

Enter the details of your location. This is used to set the language, time zone, keyboard and other international settings.

Country: United States

Language: American English

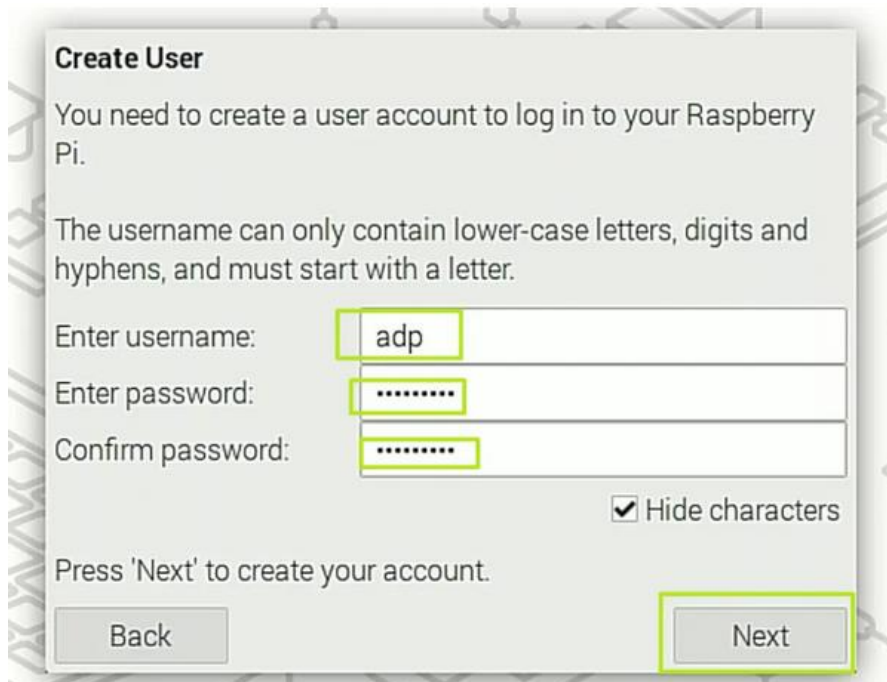
Timezone: Eastern

☒ Use English language ☒ Use US keyboard

Press 'Next' when you have made your selection.

Back Next

3. **Enter a username and password** you wish to use for your primary login. Click Next



Create User

You need to create a user account to log in to your Raspberry Pi.

The username can only contain lower-case letters, digits and hyphens, and must start with a letter.

Enter username: adp

Enter password: *****

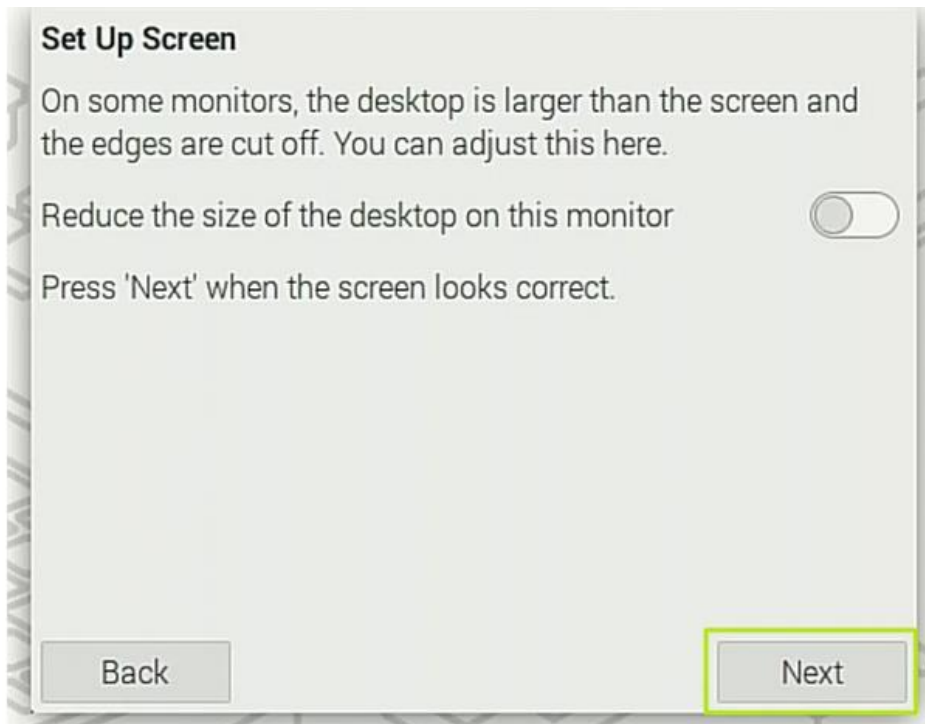
Confirm password: *****

☒ Hide characters

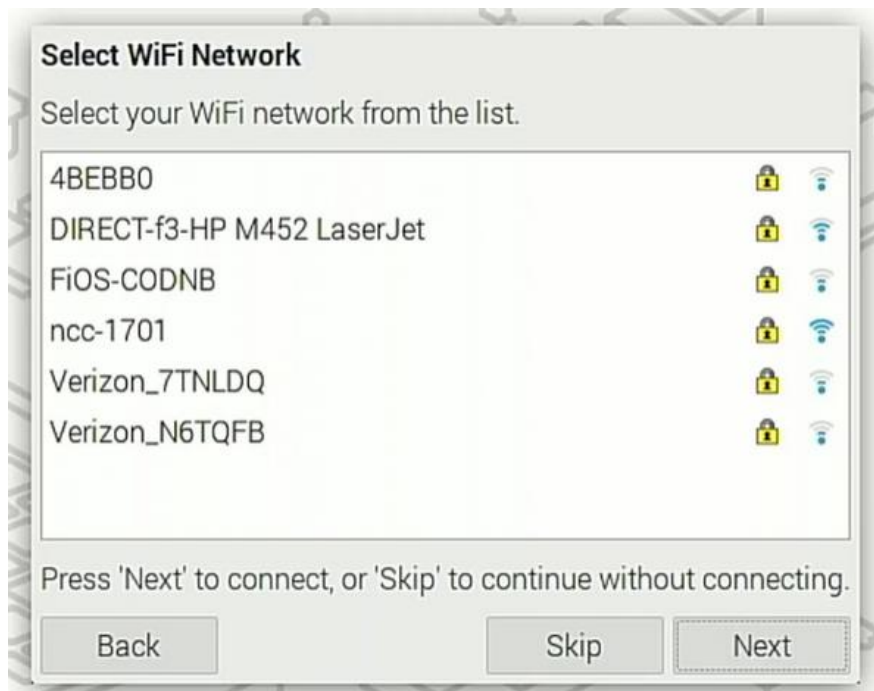
Press 'Next' to create your account.

Back Next

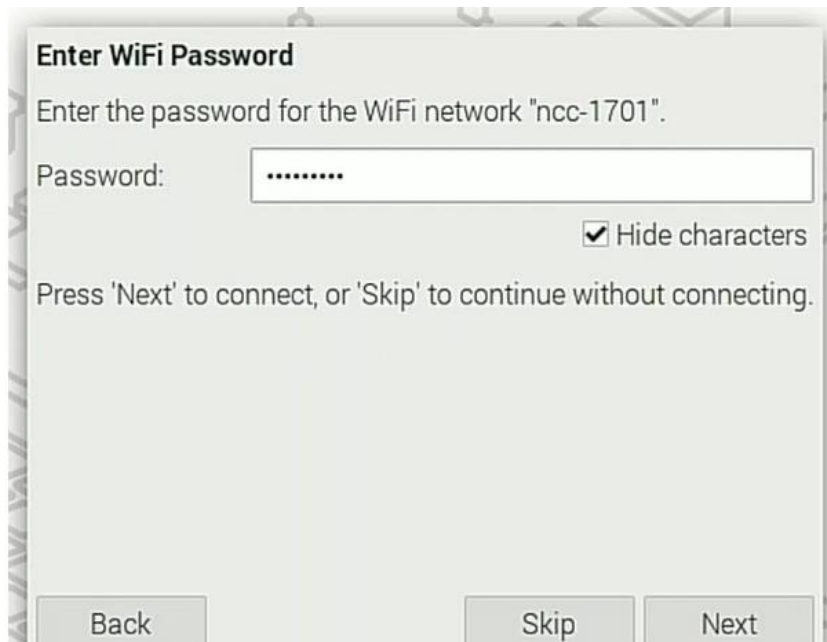
4. **Toggle "Reduce the size of the desktop" to on** if the borders of the desktop are cut off. Otherwise, just **click Next**.



5. **Select the appropriate Wi-Fi network** on the screen after, provided that you are connecting via Wi-Fi. If you don't have Wi-Fi or are using Ethernet, you can skip this.

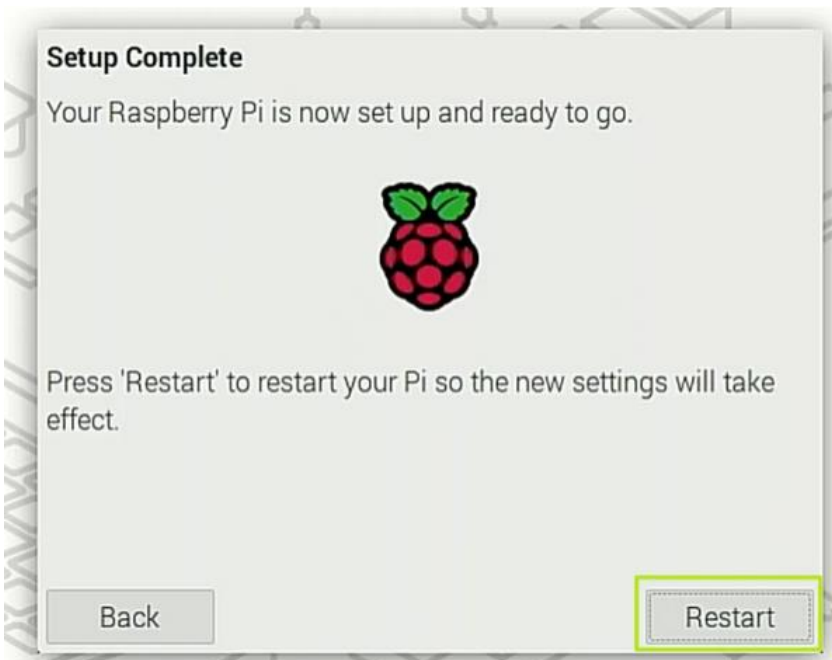


6. Enter your Wi-Fi password (unless you were using Ethernet and skipped).



The screenshot shows a dialog box titled "Enter WiFi Password". It contains the text "Enter the password for the WiFi network 'ncc-1701'." followed by a "Password:" label and a text input field with seven dots. To the right of the input field is a checkbox labeled "Hide characters" which is checked. Below the input field is the instruction "Press 'Next' to connect, or 'Skip' to continue without connecting." At the bottom of the dialog are three buttons: "Back", "Skip", and "Next".

8. Click Restart.



The screenshot shows a dialog box titled "Setup Complete". It contains the text "Your Raspberry Pi is now set up and ready to go." followed by the Raspberry Pi logo (a red raspberry with two green leaves). Below the logo is the instruction "Press 'Restart' to restart your Pi so the new settings will take effect." At the bottom of the dialog are two buttons: "Back" and "Restart". The "Restart" button is highlighted with a yellow rectangular border.

If you wish to change these settings later, you can find the region and password settings, along with many other options, by clicking on the Pi icon in

the upper left corner of the screen and navigating to *Preferences* -> *Raspberry Pi Configuration*. You can configure Wi-Fi by clicking on the Wi-Fi / network icon on the taskbar.



3.THINGSBOARD USER GUIDE

What is ThingsBoard?

ThingsBoard is an open-source IoT platform that enables rapid development, management, and scaling of IoT projects. Our goal is to provide the out-of-the-box IoT cloud or on-premises solution that will enable server-side infrastructure for your IoT applications.

Features

With ThingsBoard, you are able to:

- Provision devices, assets and customers, and define relations between them.
- Collect and visualize data from devices and assets.
- Analyze incoming telemetry and trigger alarms with complex event processing.
- Control your devices using remote procedure calls (RPC).
- Build work-flows based on a device life-cycle event, REST API event, RPC request, etc.
- Design dynamic and responsive dashboards and present device or asset telemetry and insights to your customers.
- Enable use-case specific features using customizable rule chains.
- Push device data to other systems.

Getting Started with ThingsBoard

1. Upload Data to Thingsboard

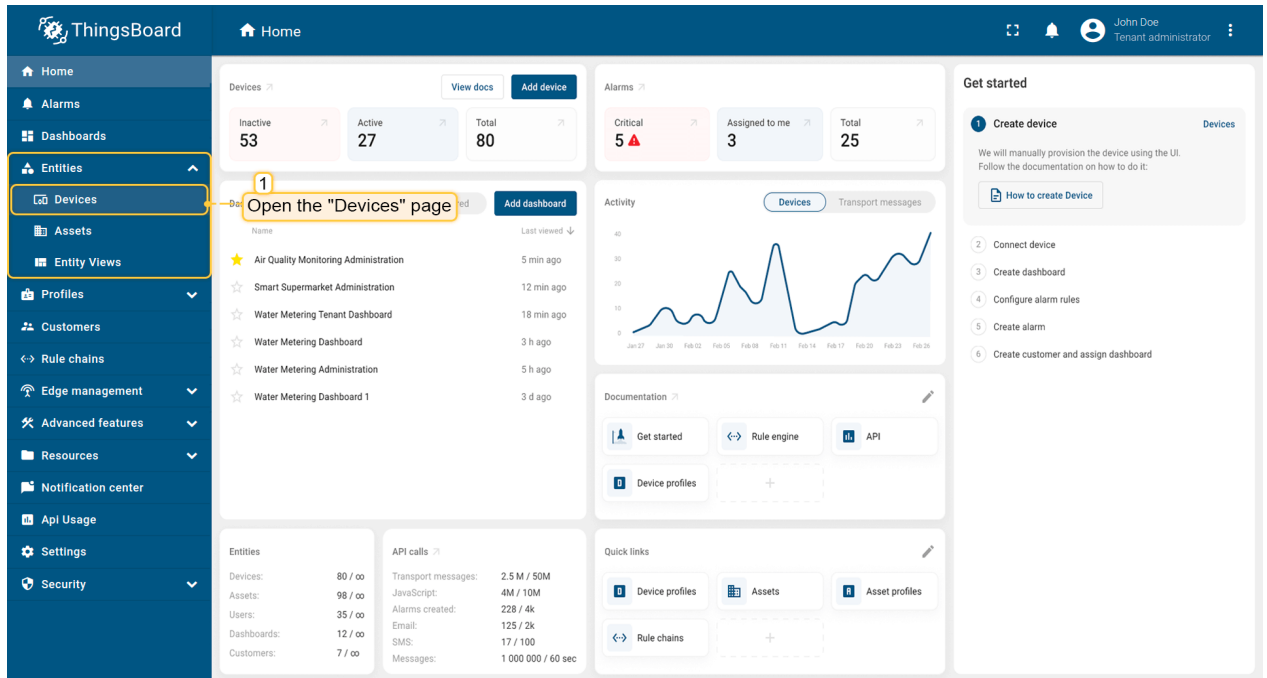
Introduction

The goal of this tutorial is to demonstrate the basic usage of the most popular ThingsBoard features. You will learn how to:

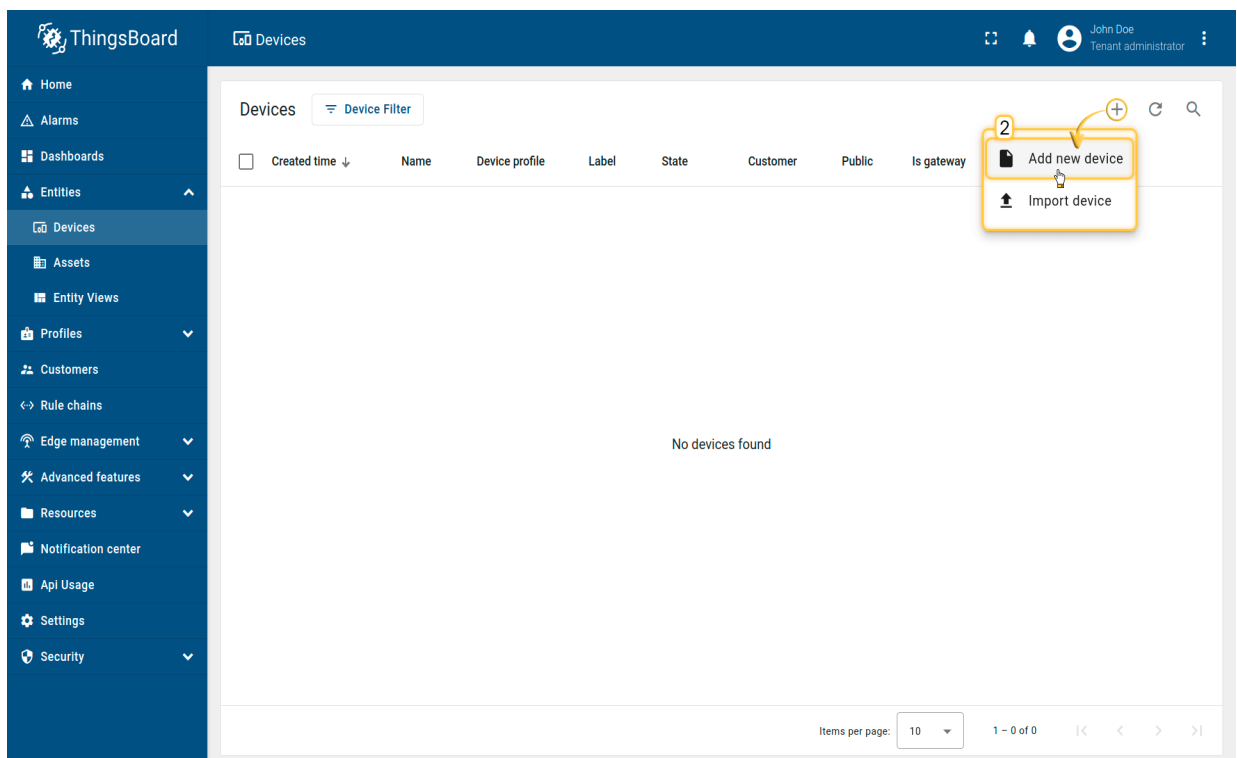
- Connect devices to ThingsBoard;
- Push data from devices to ThingsBoard;
- Build real-time end-user dashboards;
- Define thresholds and trigger alarms;
- Push notifications about new alarms over email, sms or other systems.
- We will connect and visualize data from the temperature sensor to keep it simple.

Step 1. Provision Device

- For simplicity, we will provision the device manually using the UI.
- Login to your ThingsBoard instance and navigate to the "Entities". Then click the "Devices" page;



- Click on the "+" icon in the top right corner of the table and then select "Add new device";



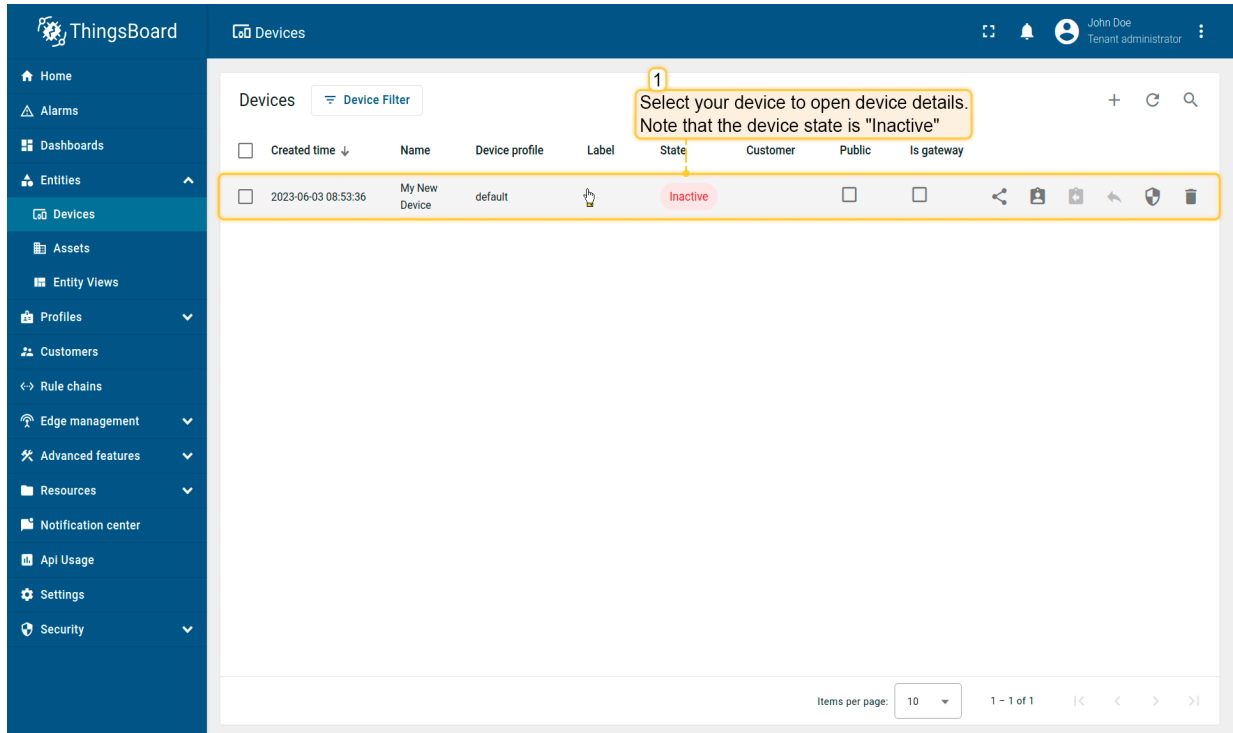
- Input device name. For example, "My New Device". No other changes required at this time. Click "Add" to add the device

- Your first device has been added. As long as you have one device. But as new devices are added, they will be added to the top of the table, since the table sort devices using the time of the creation by default
- When adding a new device, you will receive a notification. You can view it by clicking on the "bell" icon in the top right corner.

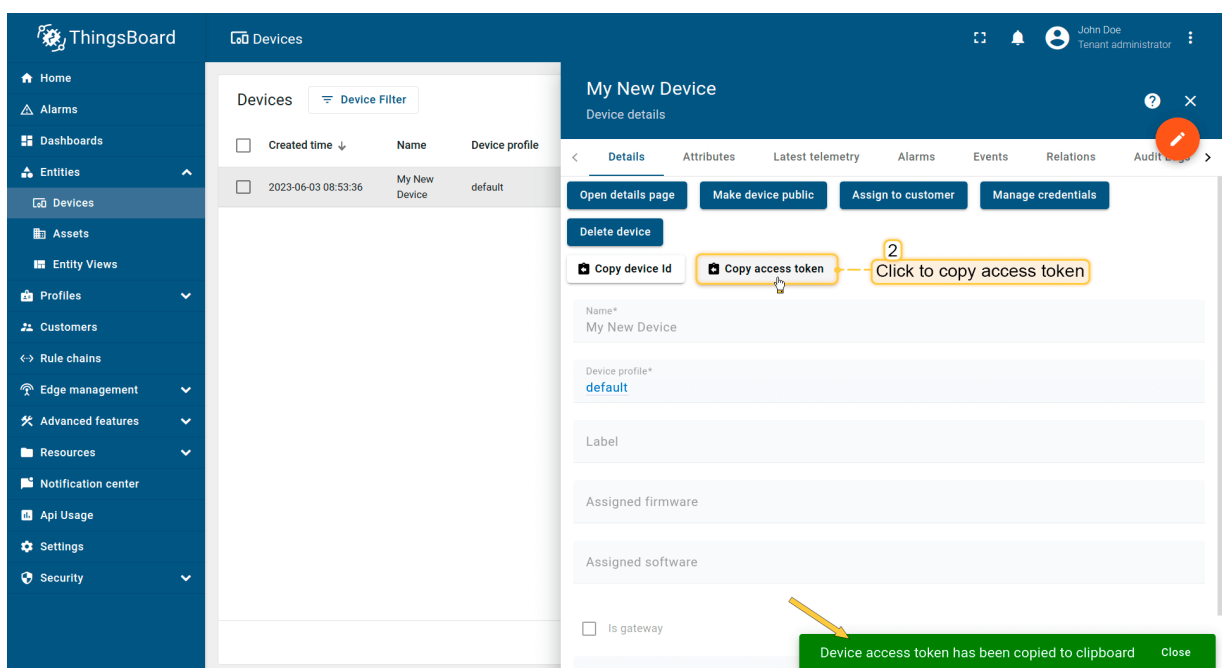
Step 2. Connect device

To connect the device you need to get the device credentials first. ThingsBoard supports various device credentials. We recommend using default auto-generated credentials which is access token for this guide.

- Click on the device row in the table to open device details. Note that the device state is "Inactive";



- Click "Copy access token". Token will be copied to your clipboard. Save it to a safe place.



Example of Publish Data of DHT22 sensor

Now you are ready to publish telemetry data on behalf of your device. We will use simple commands to publish data over MQTT in this DHT22 Sensor example. Temperature upload over MQTT using `paho-mqtt` and DHT22 sensor

ThingsBoard is an open-source server-side platform that allows you to monitor and control IoT devices. It is free for both personal and commercial usage and you can deploy it anywhere. If this is your first experience with the platform we recommend to review [what-is-thingsboard](#) page and [getting-started](#) guide.

This sample application performs collection of temperature and humidity values produced by DHT22 sensor and further visualization on the real-time web dashboard. Collected data is pushed via MQTT to ThingsBoard server for storage and visualization. The purpose of this application is to demonstrate ThingsBoard data collection API and visualization capabilities.

The DHT22 sensor is connected to `raspberrypi`. `raspberrypi` offers a complete and self-contained Wi-Fi networking solution. `raspberrypi` push data to ThingsBoard server via MQTT protocol by using `paho-mqtt` python library. Data is visualized using built-in customizable dashboard. The application that is running on `raspberrypi` is written in Python which is quite simple and easy to understand.

List of hardware and pin-outs

- Raspberry Pi 4
- DHT22 sensor

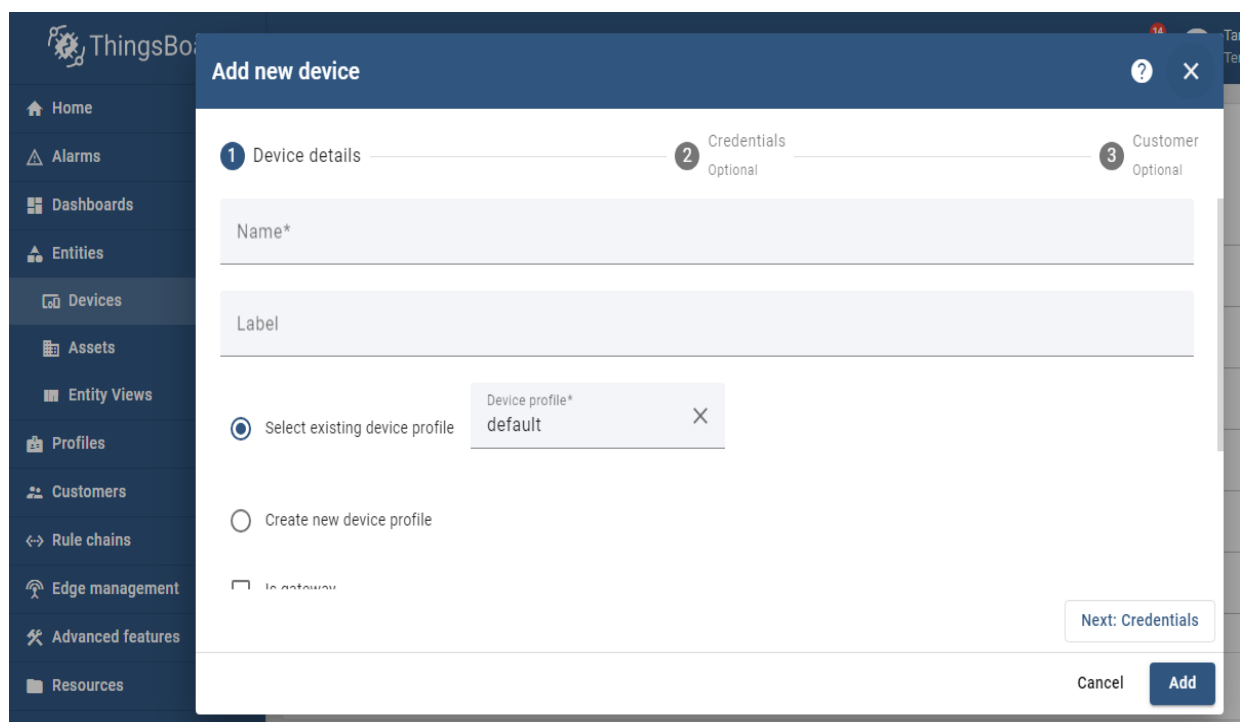
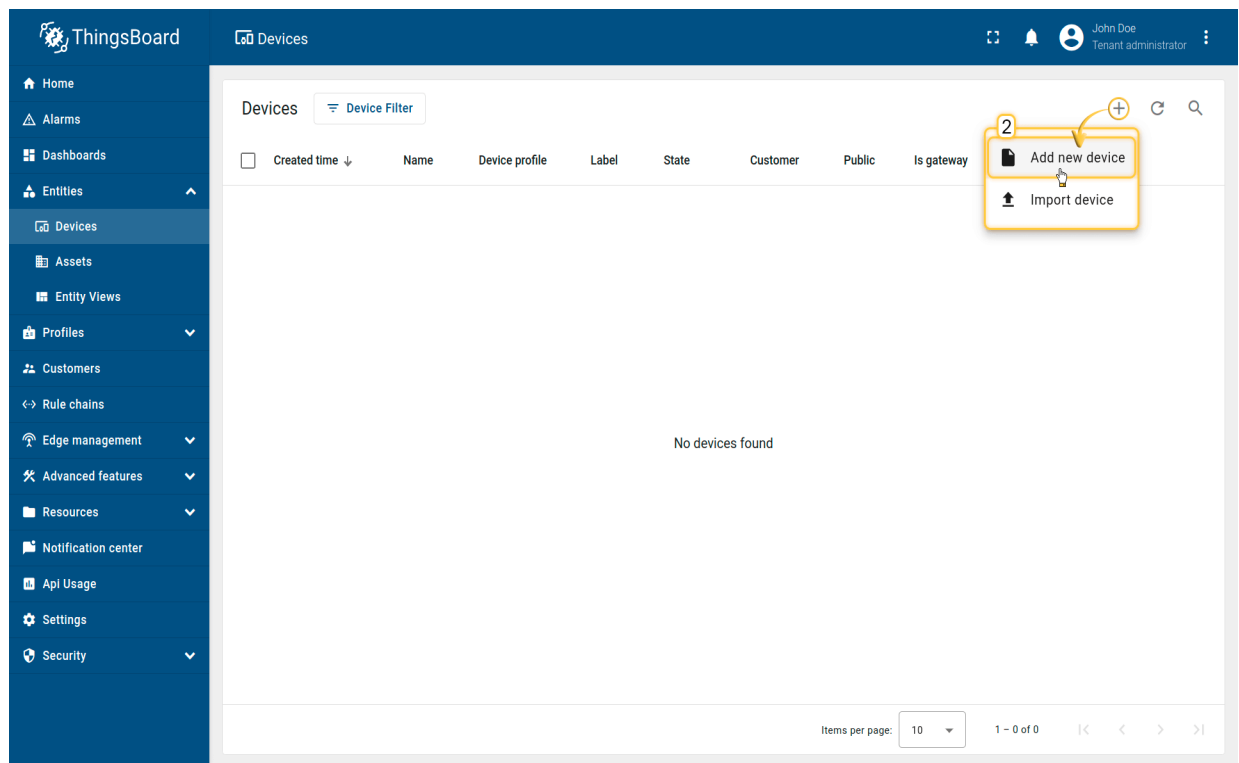
Things-Board configuration

Note Things-Board configuration steps are necessary only in case of **local Things-Board installation**. If you are using **Live Demo** instance all entities are pre-configured for your demo account. However, we recommend reviewing this steps because you will still need to get device access token to send requests to Things-Board.

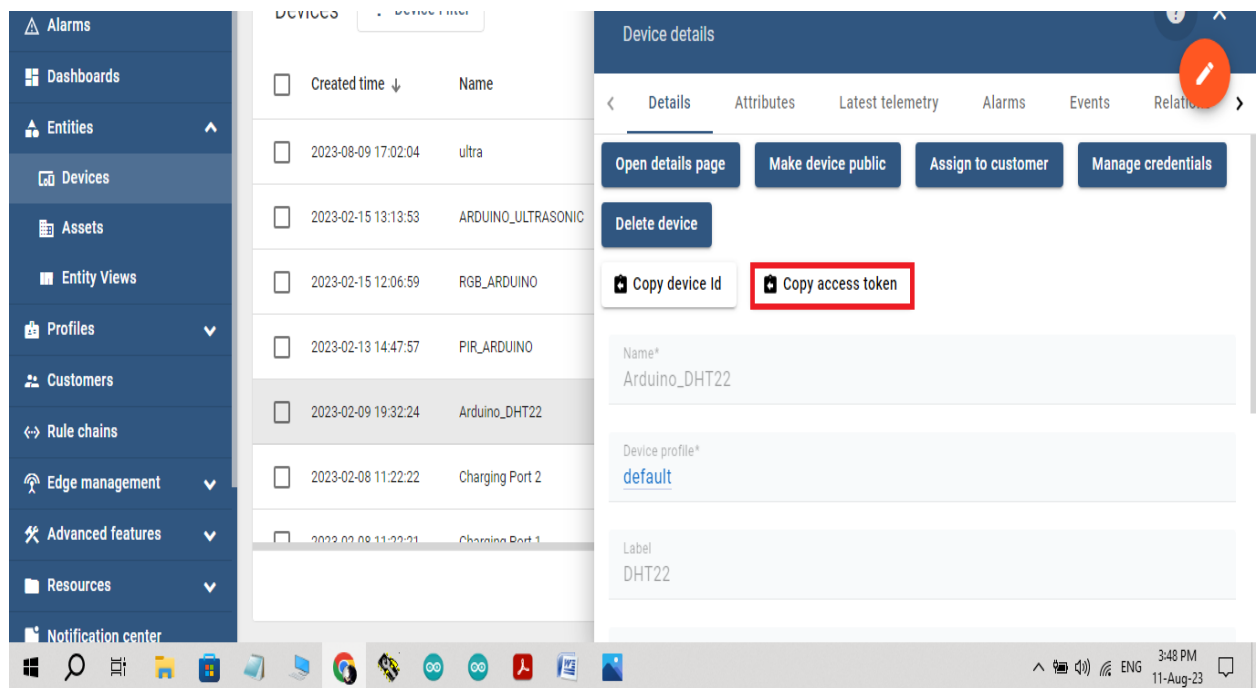
Provision your device

This step contains instructions that are necessary to connect your device to ThingsBoard.

- Open ThingsBoard in browser and login
- Goto “Devices” section. Click “+” button and create a device with the name “DHT22 Demo Device”.



Once device created, open its details and click “Manage credentials”. Copy auto-generated access token from the “Access token” field. Please save this device token. It will be referred to later as **\$ACCESS_TOKEN**.



Click “Copy access token” in device details to copy your device id to the clipboard. Paste your device id to some place, this value will be used in further steps.

Programming the

- Power on the
- MQTT library installation

The following command will install MQTT Python library:

- **sudo pip install paho-mqtt**
- **Adafruit DHT library installation**

Install python-dev package:

- **sudo apt-get install python-dev**

Downloading and install the Adafruit DHT library:

- **git clone https://github.com/adafruit/Adafruit_Python_DHT.git**
- **cd Adafruit_Python_DHT**
- **sudo python setup.py install**
- **Goto → home/logsun/Documents/Things-board**
- **Right click on dht22 code and open in Thonny IDE**



Replace your access token with your copied access token previously you have done.

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mqtt-dht22.py

```

1 import os
2 import time
3 import sys
4 import Adafruit_DHT as dht
5 import paho.mqtt.client as mqtt
6 import json
7
8 THINGSBOARD_HOST = 'demo.thingsboard.io'
9 ACCESS_TOKEN = 'DHT22_DEMO_TOKEN'
10
11 # Data capture and upload interval in seconds. Less interval will eventu
12 INTERVAL=2
13
14 sensor_data = {'temperature': 0, 'humidity': 0}

```

- Provision your dashboard
- Programming the Raspberry Pi
 - MQTT library installation
 - Adafruit DHT library installation
 - Application source code
- Running the application
- Data visualization
- See also
- Your feedback
- Next steps

- Click on Run to run the program

Data visualization

Finally, open ThingsBoard You can access this dashboard by logging

Go to “**Devices**” section and locate “**DHT22 Demo Device**”, open device details and switch to “**Latest telemetry**” tab. If all is configured correctly you should be able to see latest values of “*temperature*” and “*humidity*” in the table.

DHT22 DEMO DEVICE

Device details

?

×

DETAILS

ATTRIBUTES

LATEST TELEMETRY

EVENTS

Latest telemetry

☐

Last update time

Key ↑

Value

☐

2017-01-04 19:54:01

humidity

34.1

☐

2017-01-04 19:54:01

temperature

23.0

Page:

1 ▾

Rows per page:

5 ▾

1 - 2 of 2

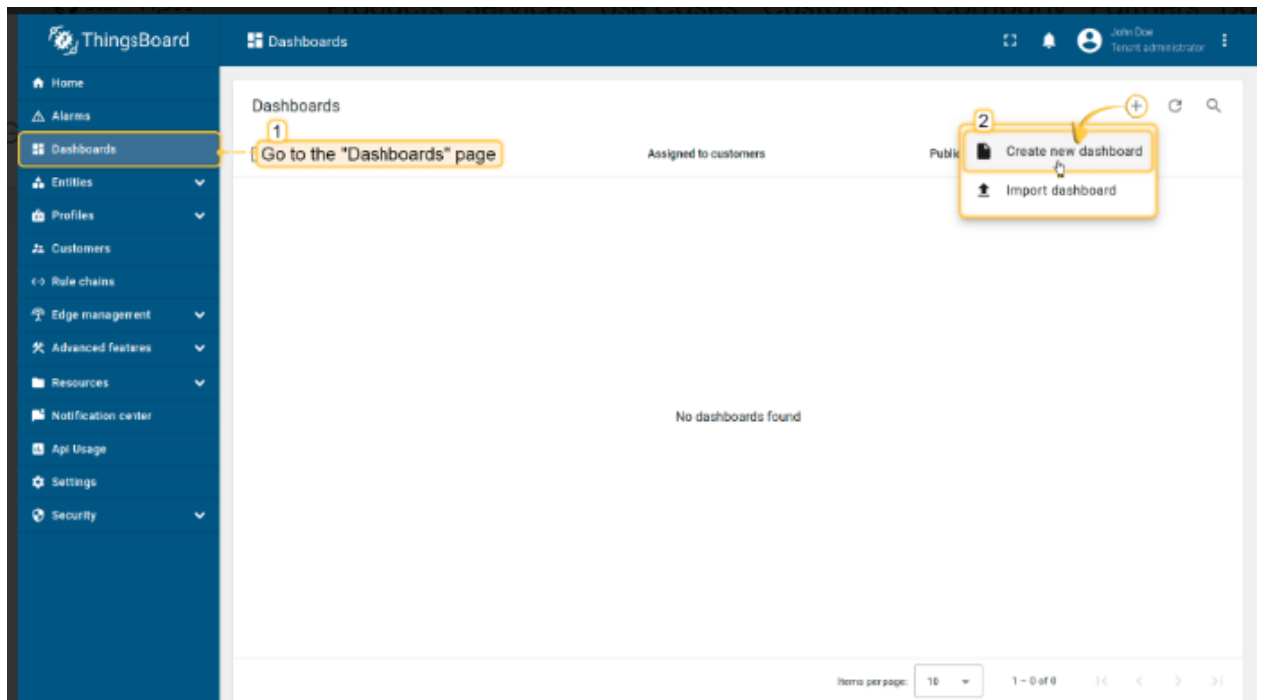
<

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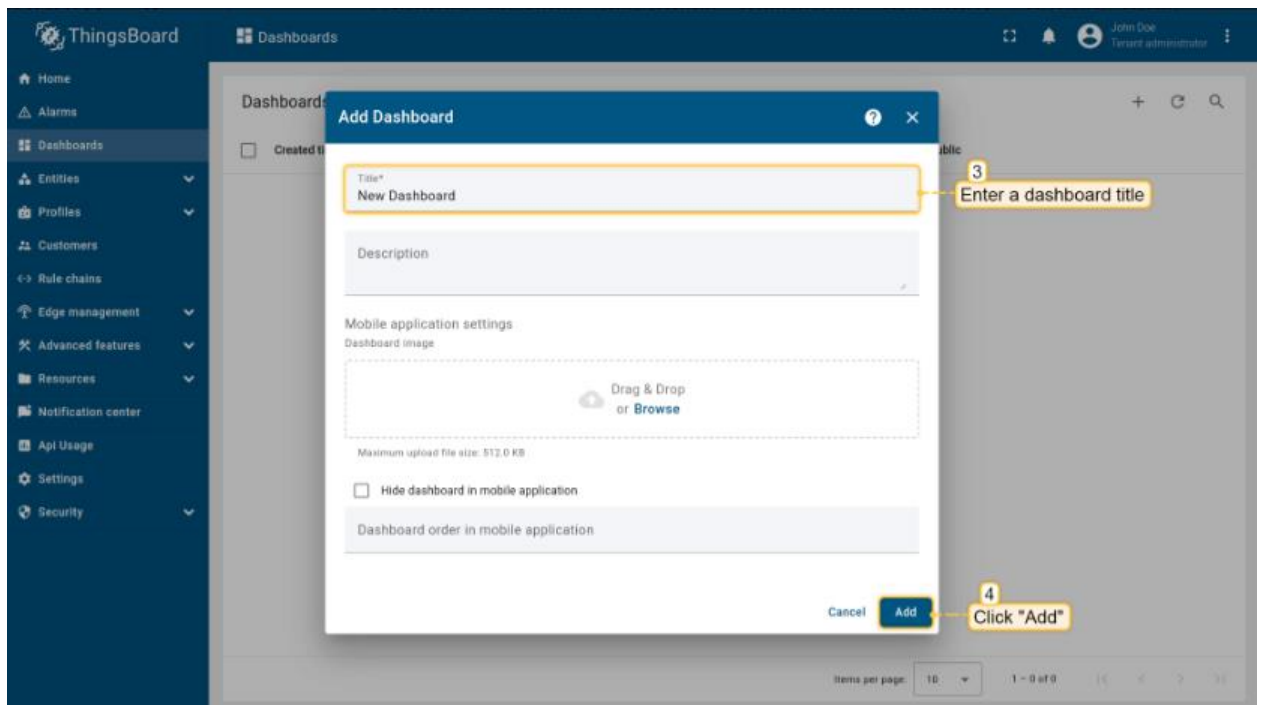
Create new dashboard

To add a new dashboard, you should:

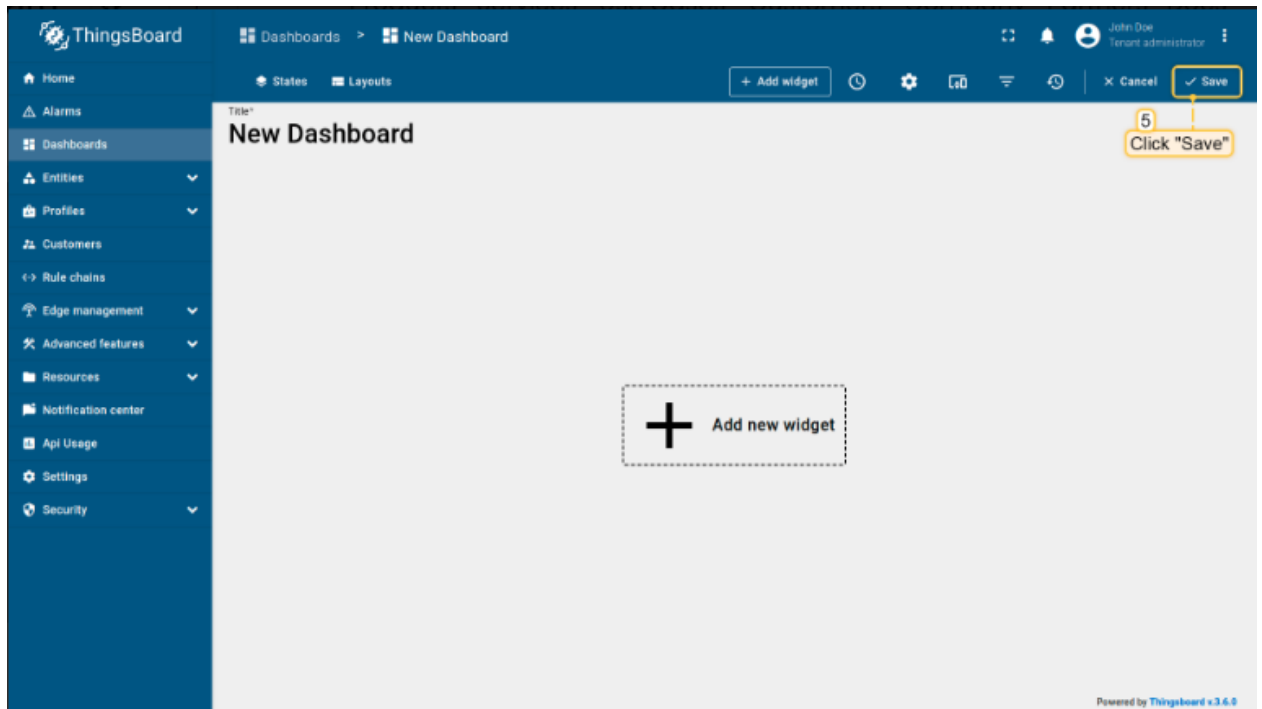
- Login to your ThingsBoard instance and navigate to the "Dashboards" page through the main menu on the left of the screen. Then, click the "+" sign in the upper right corner of the screen, and select "Create new dashboard" from the drop-down menu



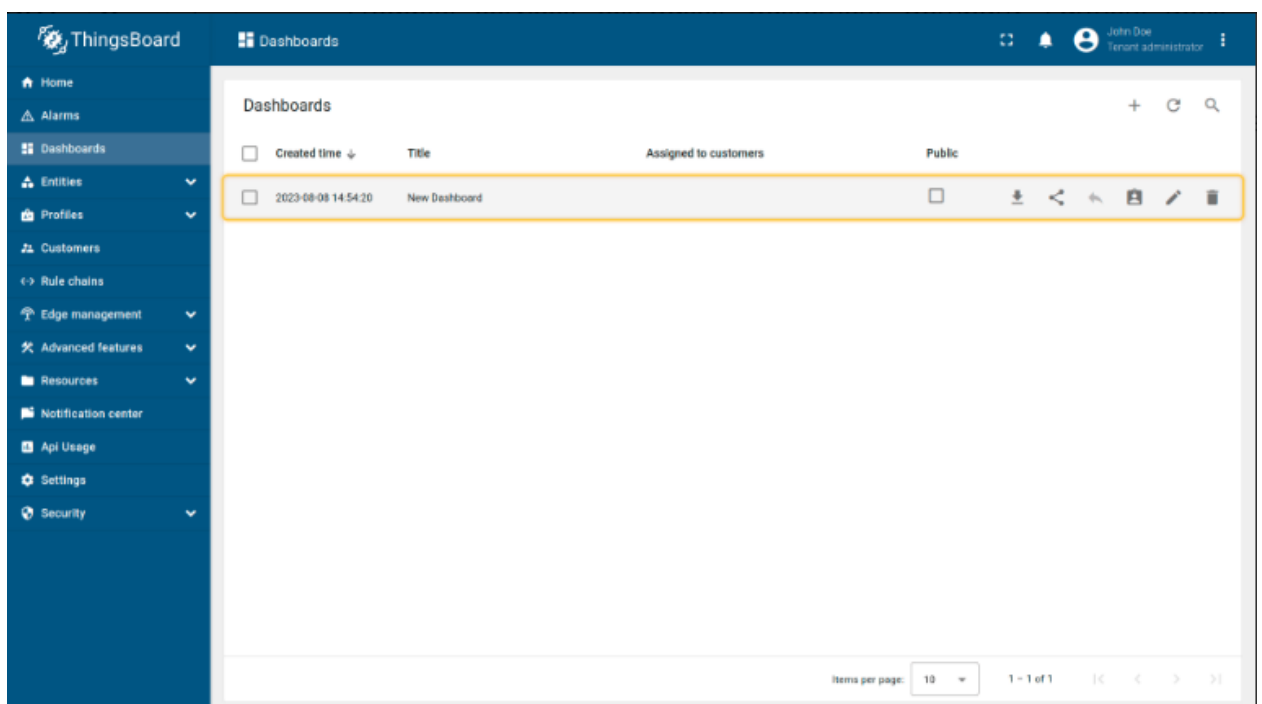
- In the opened dialog, it is necessary to enter a dashboard title, description is optional. Click "Add";



- Once you have created the dashboard, it will be automatically opened. Save it by clicking the "Save" button in the upper right corner.



- Your first dashboard has been successfully created. As you continue to add new dashboards, they will appear at the top of the list. This default sort is based on the creation timestamp.

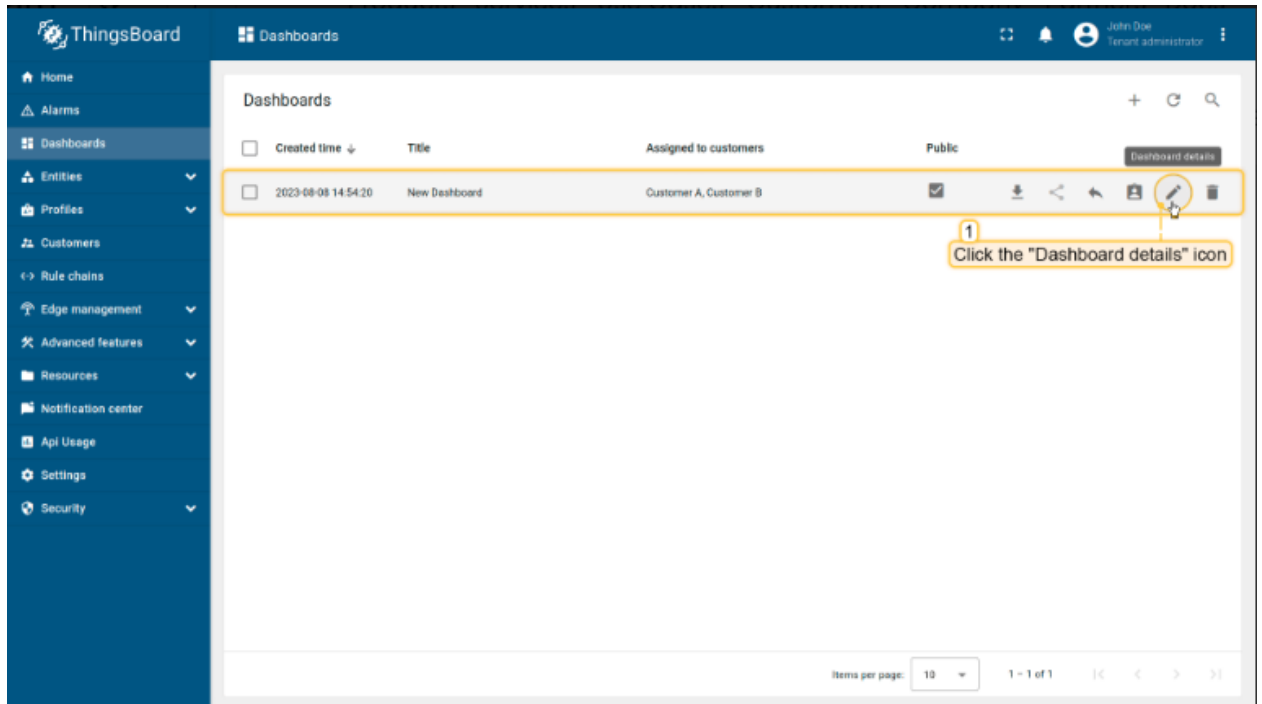


Edit dashboard

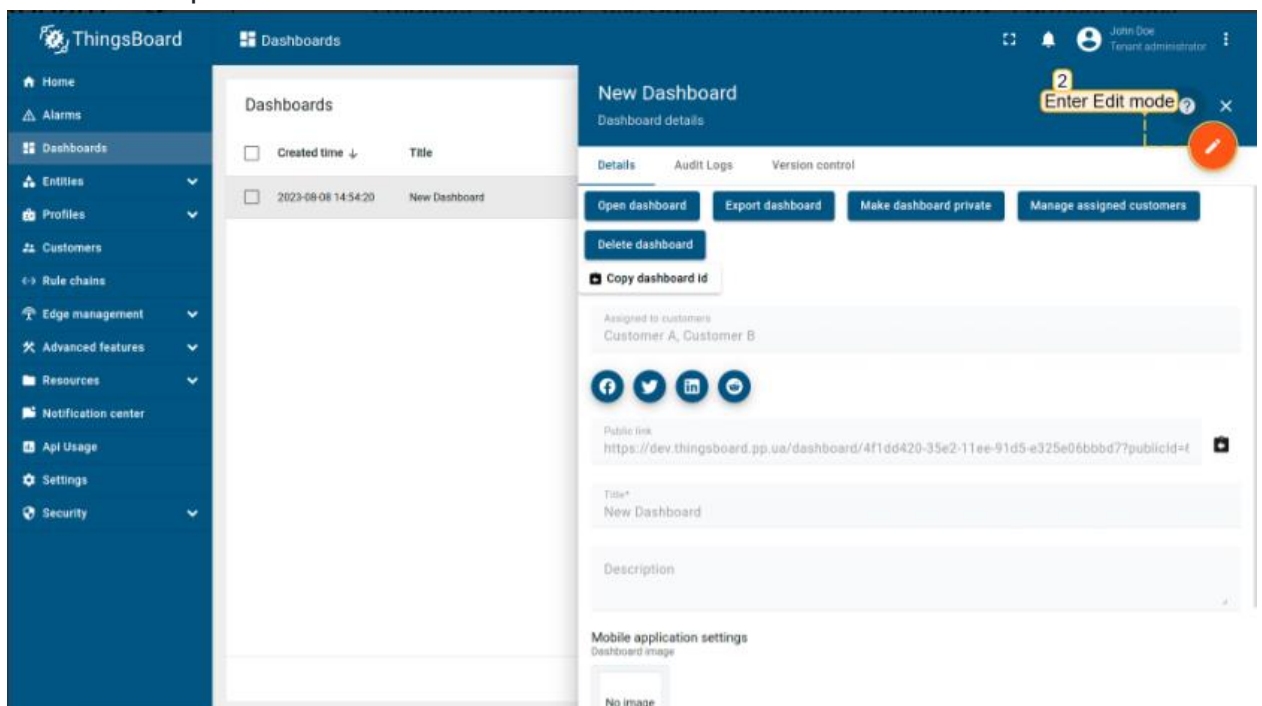
Edit the fields you need, such as the title or description of the dashboard. After making the edits, remember to save all changes. With these steps, you will have successfully updated the dashboard information.

Let's see how to do this:

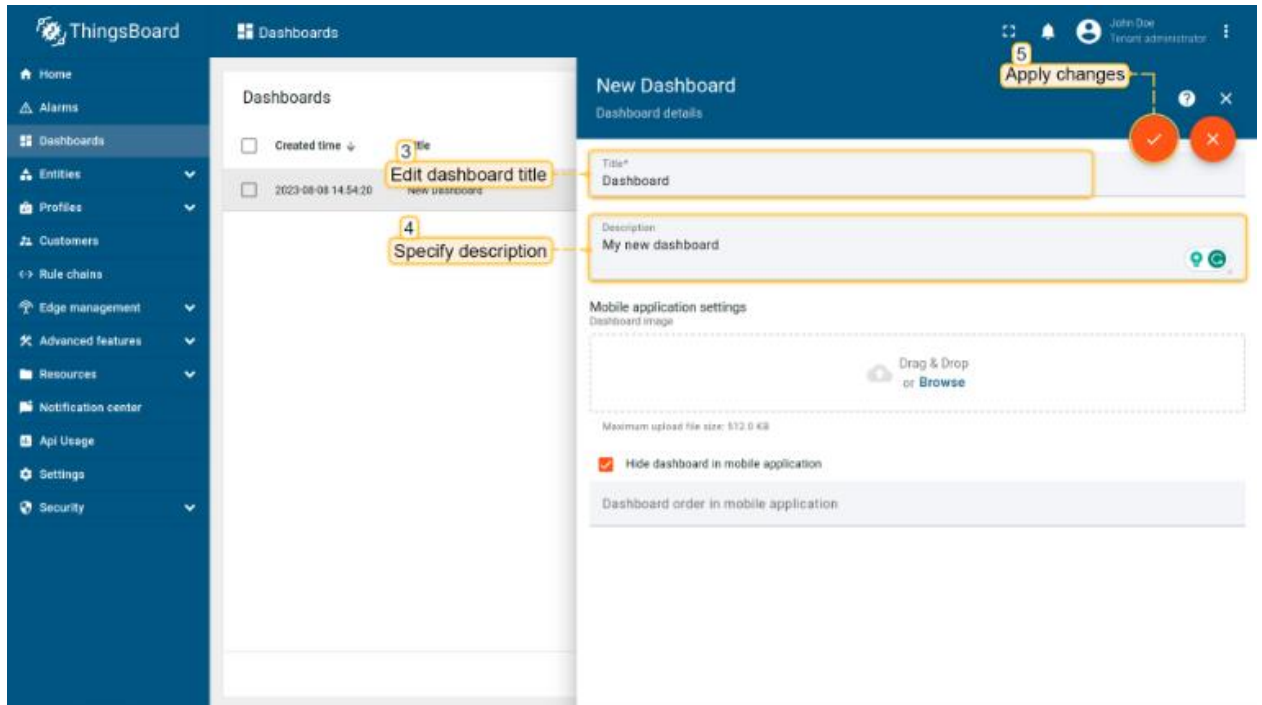
- Go to the "Dashboards" page and click the "Dashboard details" icon next to the dashboard that you want to edit



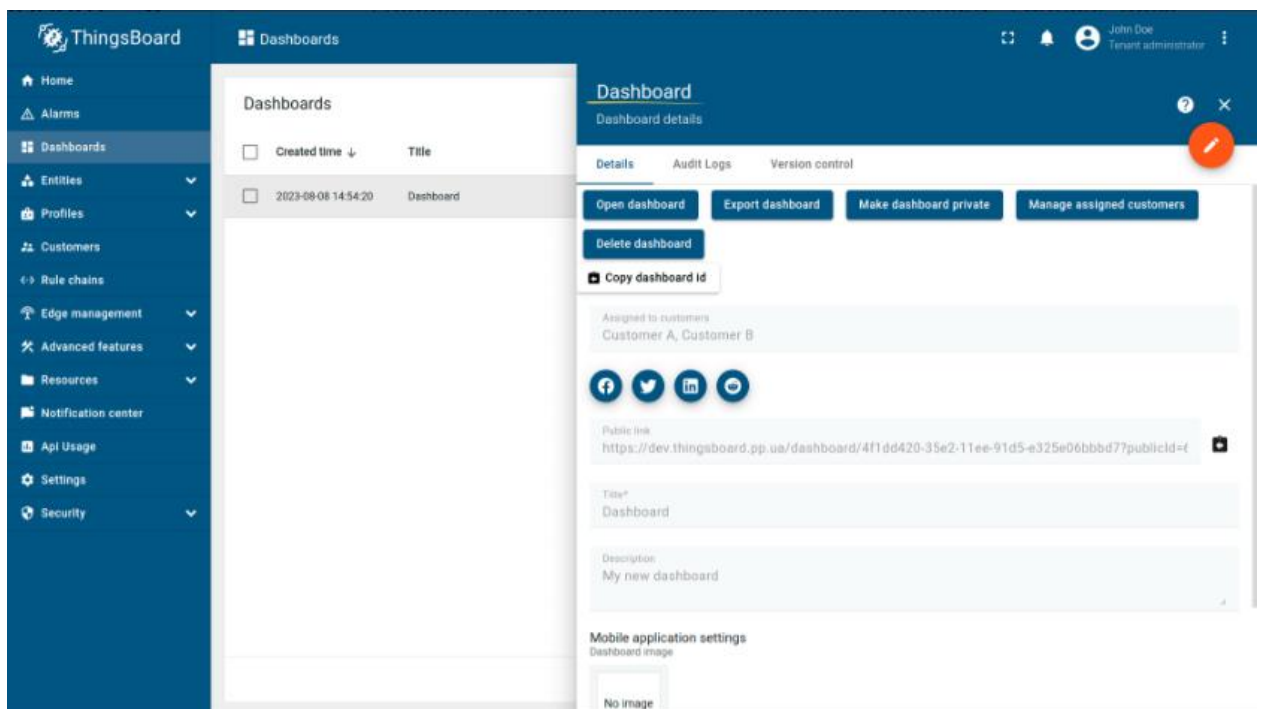
- Click the "pencil" icon to enter edit mode



- Edit the fields you need. For example, edit the title of the dashboard, specify description. After making the edits, remember to save all changes



- You have updated the dashboard information.



Widgets

Introduction

All IoT dashboards are constructed using ThingsBoard widgets.

A widget is an element that displays a specific type of information or functionality on a dashboard. Widgets are used to display data and visualize information obtained from devices connected to the ThingsBoard platform, remote device control, alarms management, and display static custom HTML content.

Widget types

There are five types of widgets:

- **Timeseries** widgets display data for a specific time window. It can be either Realtime - the dynamically changed time frame for a certain latest interval, or History - a fixed historical time frame. Examples of timeseries widgets are chart widgets. Obviously, timeseries widgets are designed to display time series and not attributes;
- **Latest values** widgets display the latest values of particular attribute or time series keys. For example, device model or latest temperature reading;
- **Control** widgets allow you to send RPC commands to your devices. For example, control desired temperature on the thermostat device;
- **Alarm** widgets allow you to display alarms;
- **Static** widgets are designed to display static data. For example, floor plan or static company information.

More about widget types you can learn here.

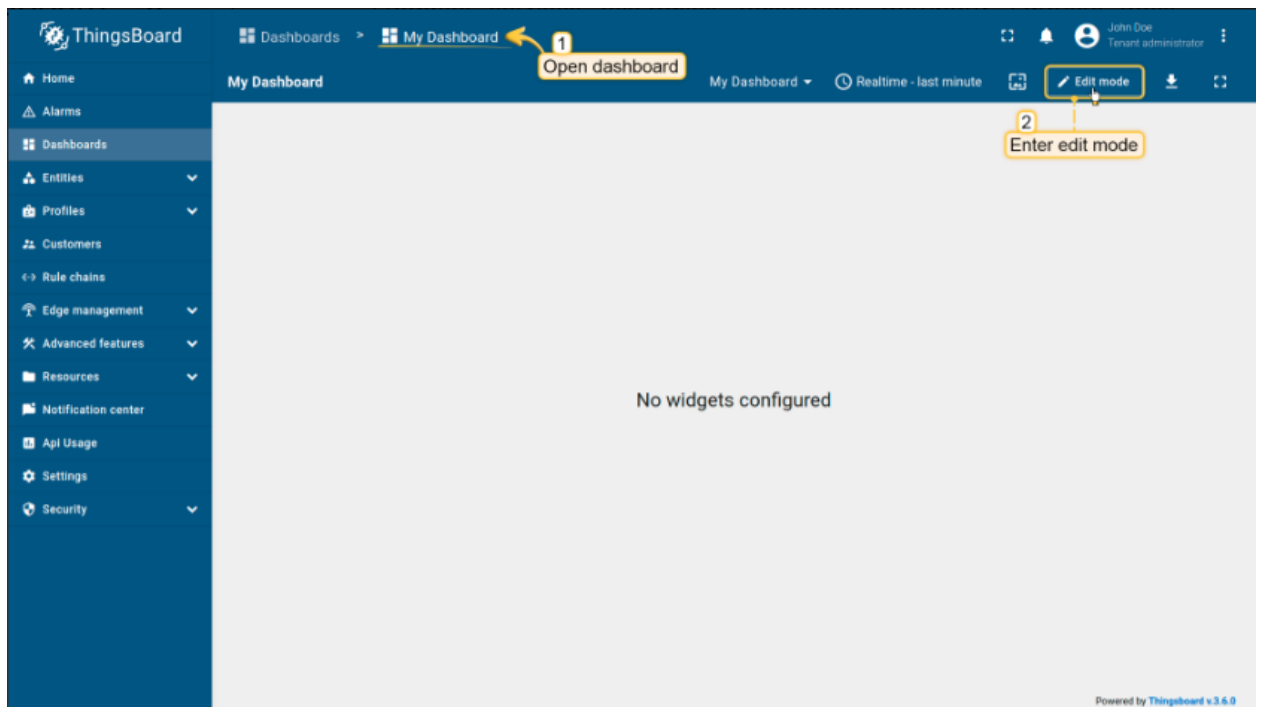
Each widget typically has specific settings and parameters that allow users to customize its behavior and appearance according to their needs.

This guide covers main concepts and various widget settings.

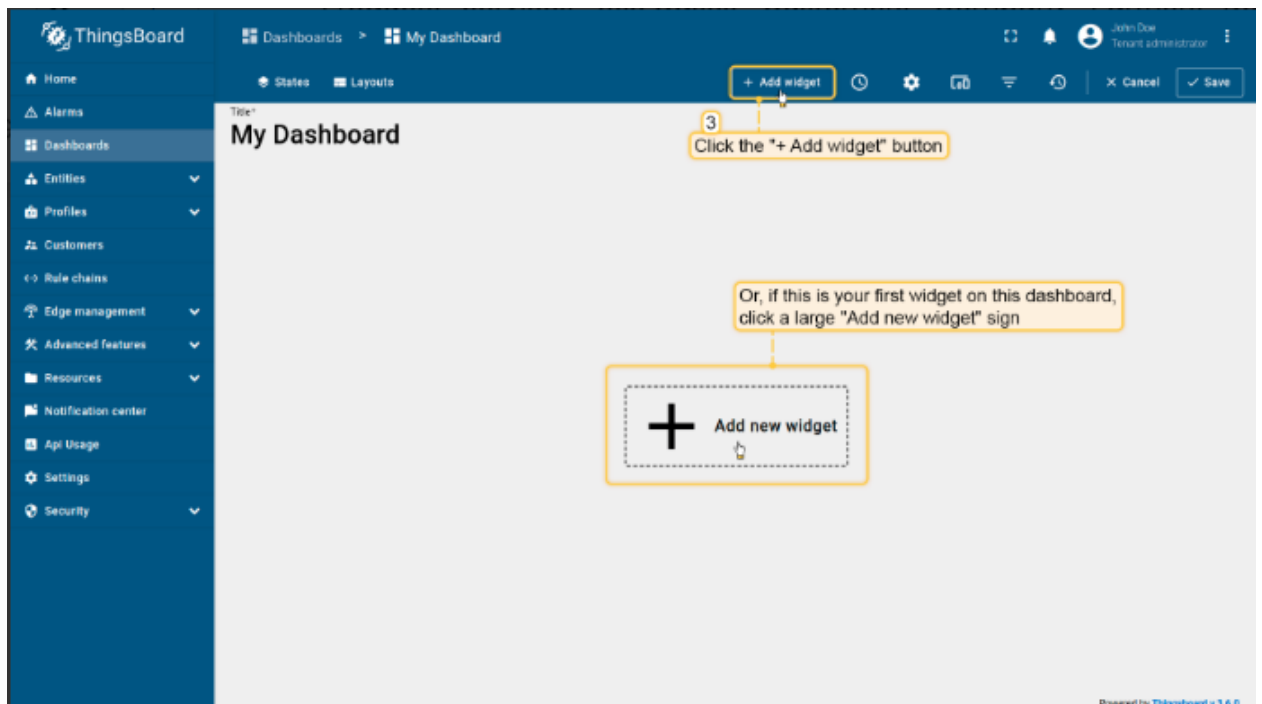
Adding a widget to the dashboard

To add a new widget to a dashboard, you should:

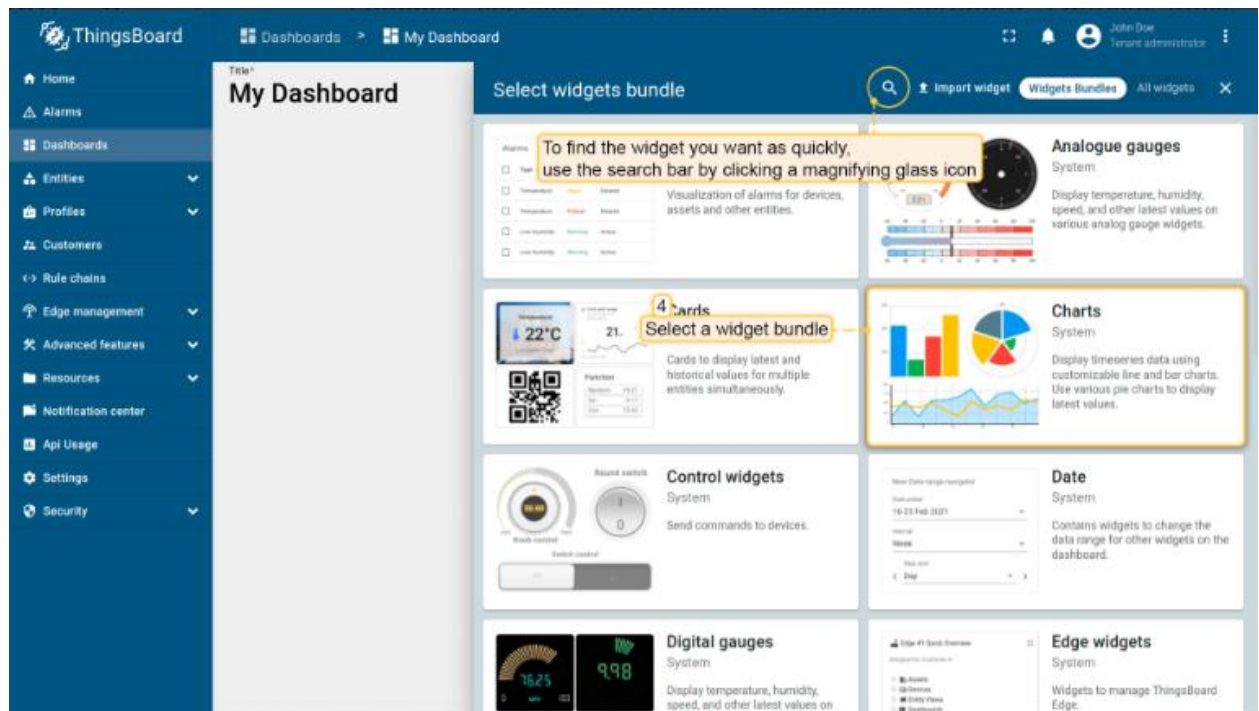
- Open your dashboard and enter edit mode



- Click the "+ Add widget" icon at the top of the screen, or (if this is your first widget on this dashboard) click a large "Add new widget" sign in the middle of the screen to open the Select widgets bundle dialog window



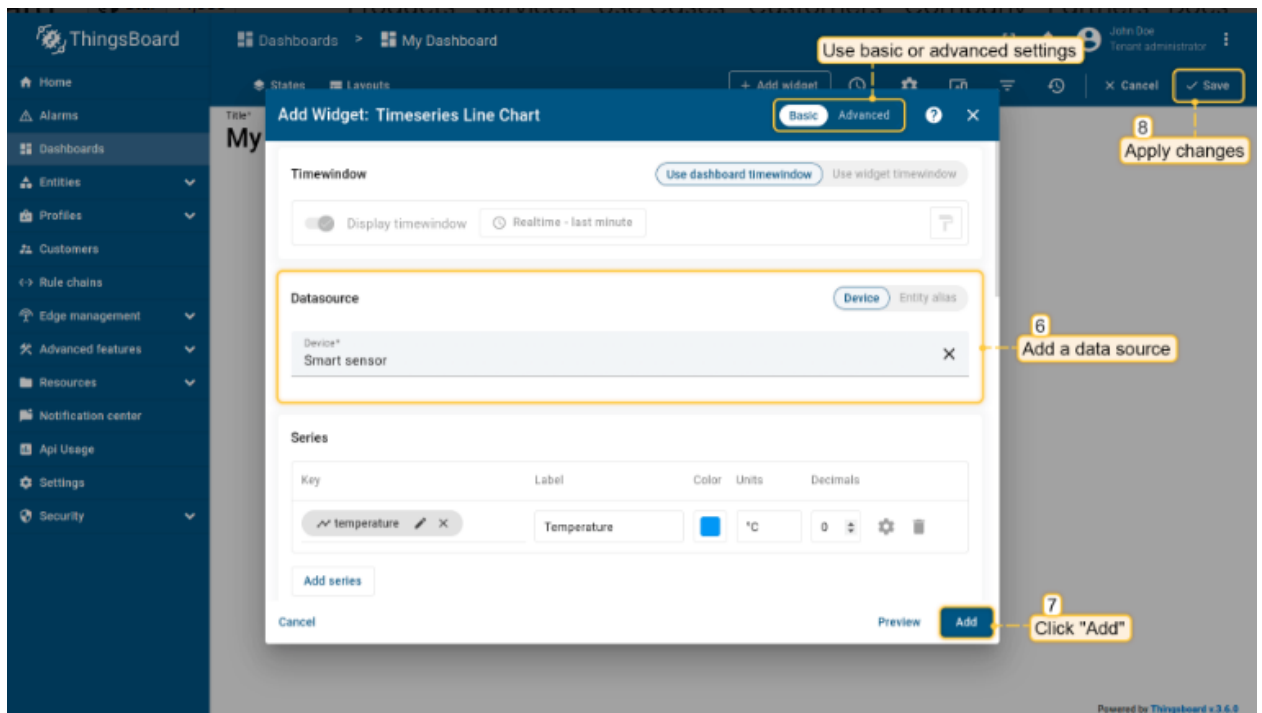
- Select a widget bundle, such as "Charts". To quickly find your desired widget, use the search bar by clicking the magnifying glass icon and entering the widget's name



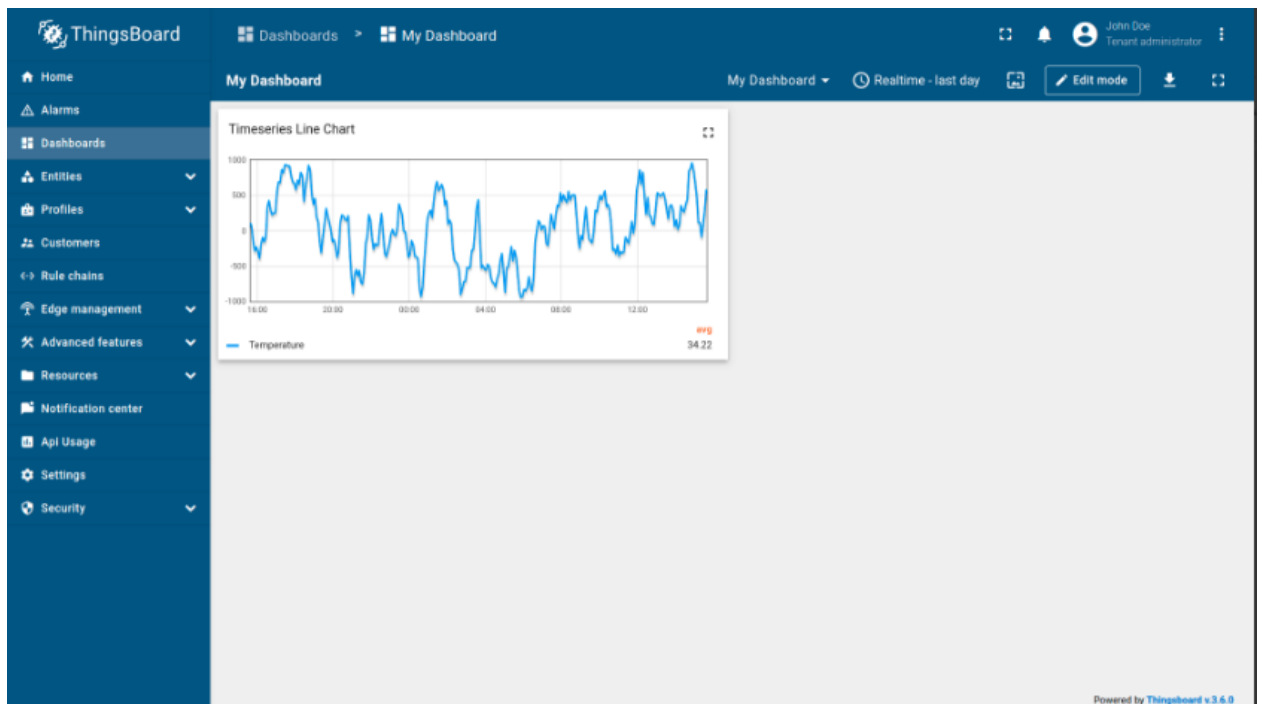
- Choose a widget, for instance, "Timeseries Line Chart" and click on it to open the "Add Widget" dialog window



- Specify the data source, add the data key, and click the "Add" button. Finally, apply your changes



- Your first widget has been created.



GPIO control over MQTT using ThingsBoard

ThingsBoard is an open-source server-side platform that allows you to monitor and control IoT devices. It is free for both personal and commercial usage and you can deploy it anywhere. If this is your first experience with the platform we recommend to review [what-is-thingsboard](#) page and [getting-started](#) guide.

This sample application will allow you to control GPIO of your device using ThingsBoard web UI. We will observe GPIO control using Led connected to one of the pins. The purpose of this application is to demonstrate ThingsBoard [RPC capabilities](#).

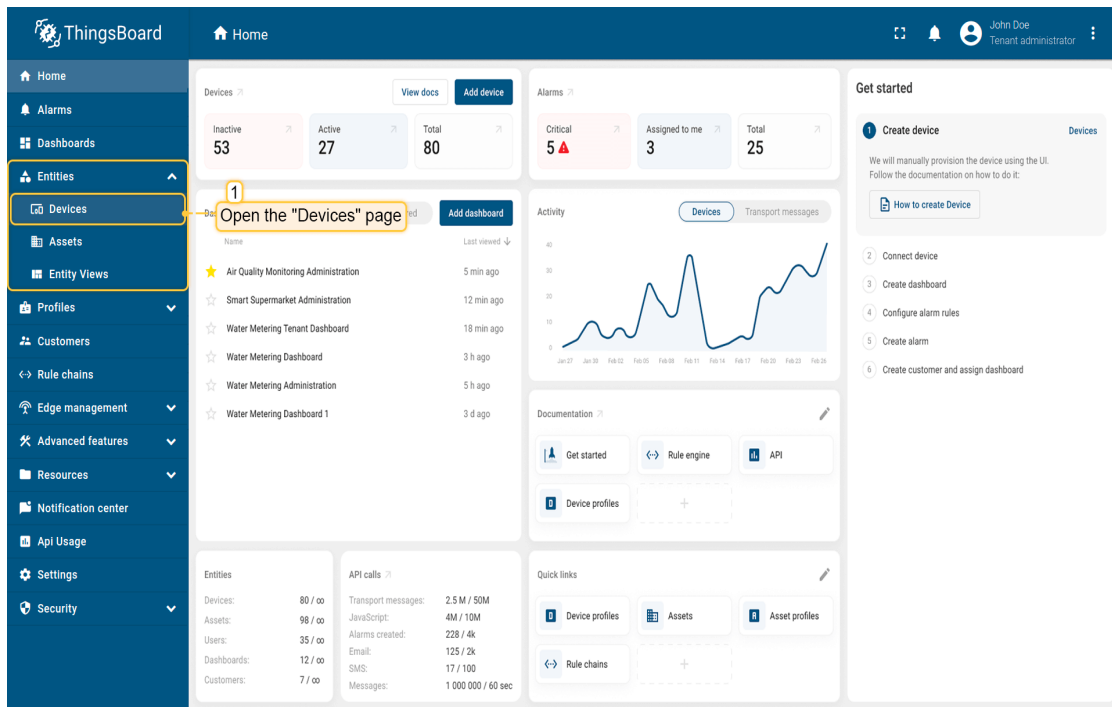
will use simple application written in Python that will connect to ThingsBoard server via [MQTT](#) and listen to RPC commands. Current GPIO state and GPIO control widget is visualized using built-in customizable dashboard.

List of hardware and pinouts

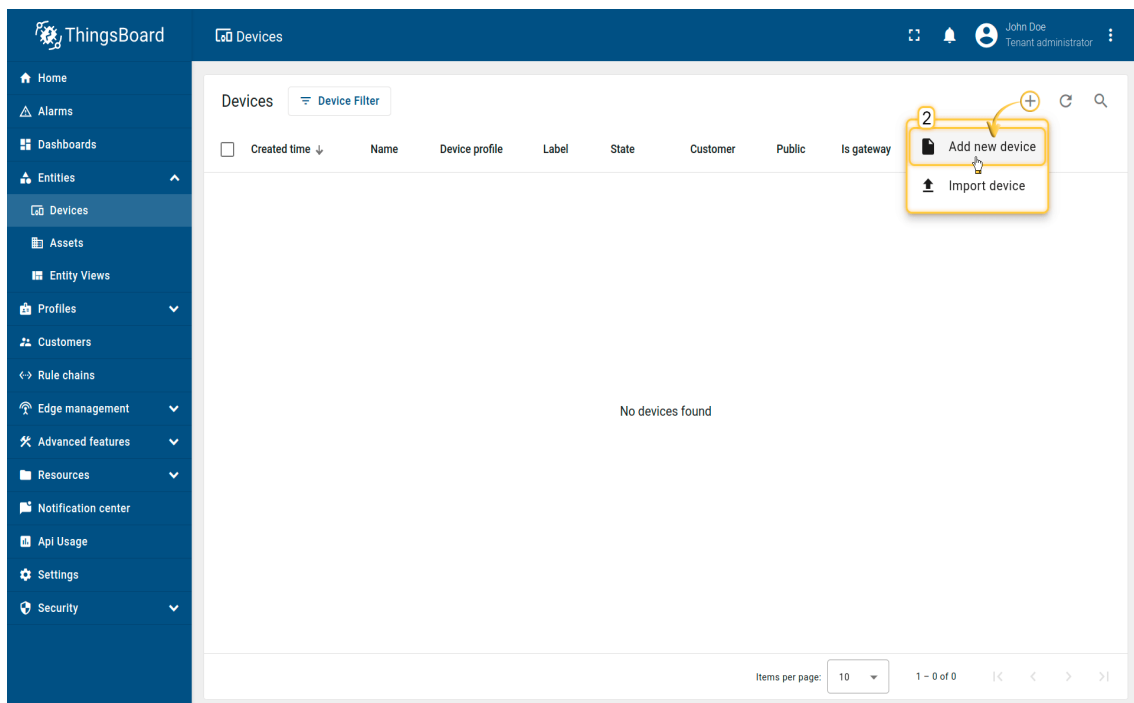
- - we will use 3 Model B but you can use any other model.
- Led and corresponding resistor

Step 1. Provision Device

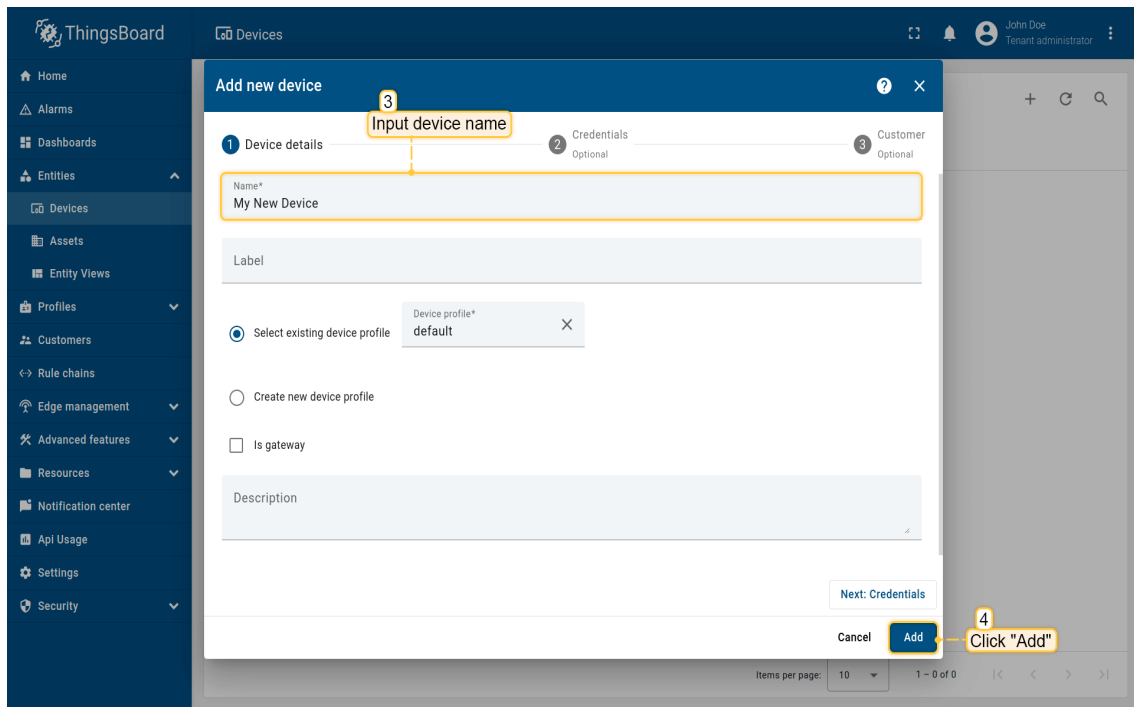
- For simplicity, we will provision the device manually using the UI.
- Login to your ThingsBoard instance and navigate to the "Entities". Then click the "Devices" page;



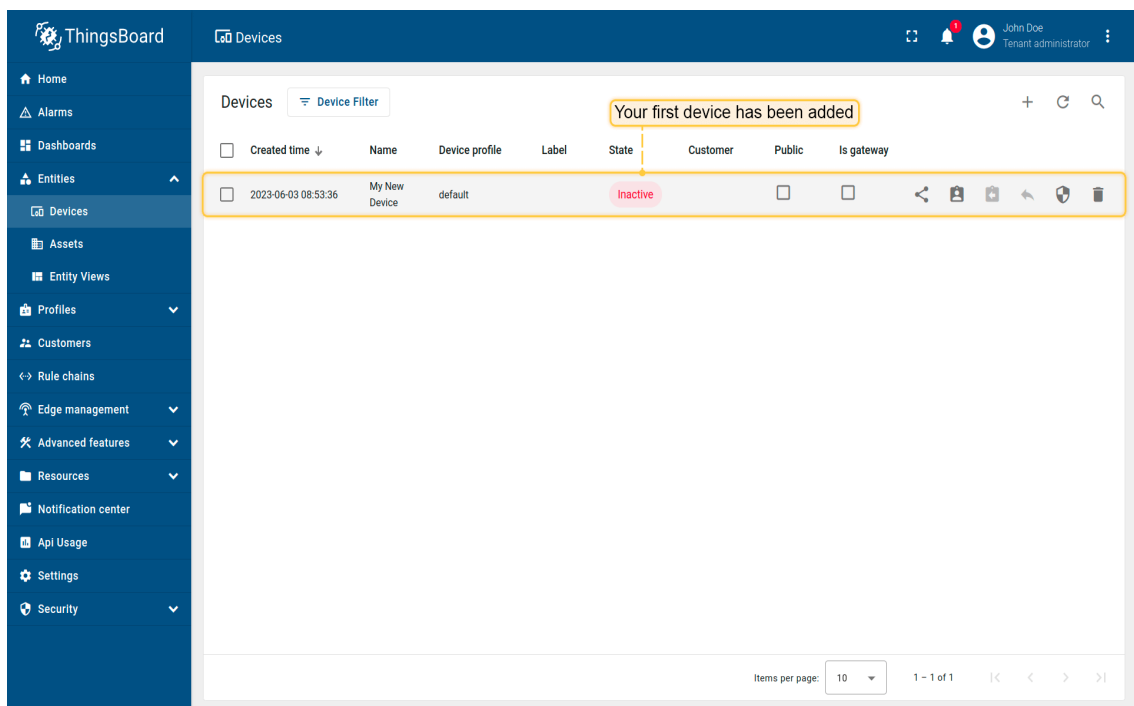
- Click on the "+" icon in the top right corner of the table and then select "Add new device";



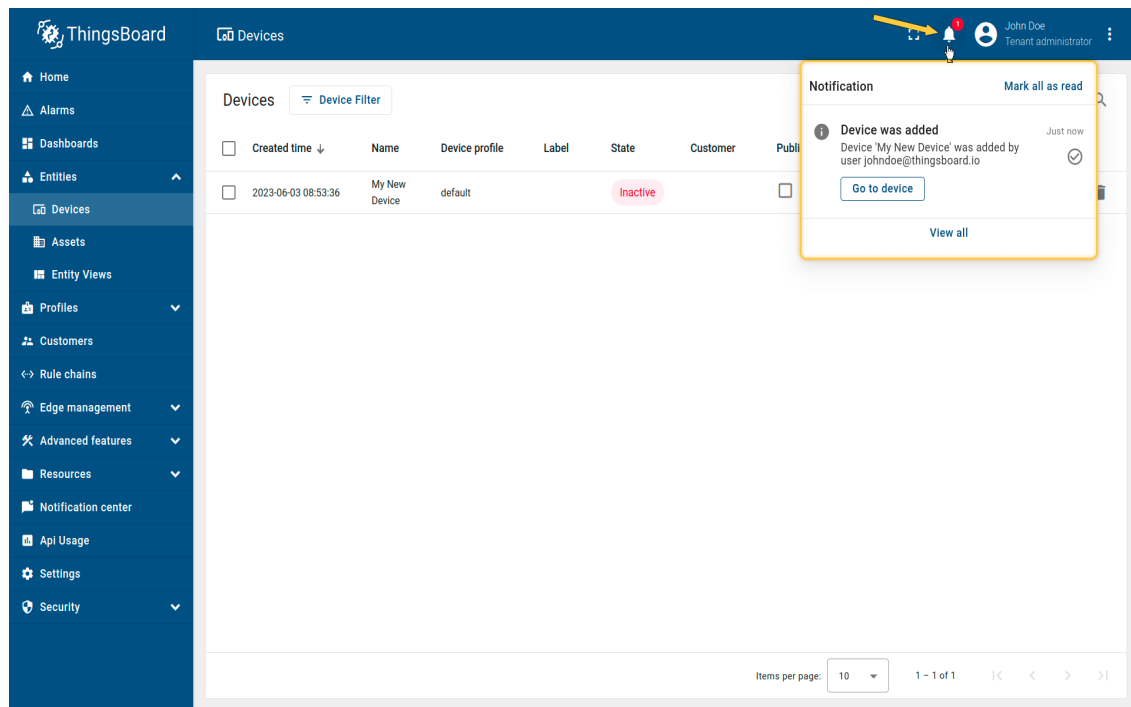
- Input device name. For example, "My New Device". No other changes required at this time. Click "Add" to add the device;



- Your first device has been added. As long as you have one device. But as new devices are added, they will be added to the top of the table, since the table sort devices using the time of the creation by default;



- When adding a new device, you will receive a notification. You can view it by clicking on the "bell" icon in the top right corner.

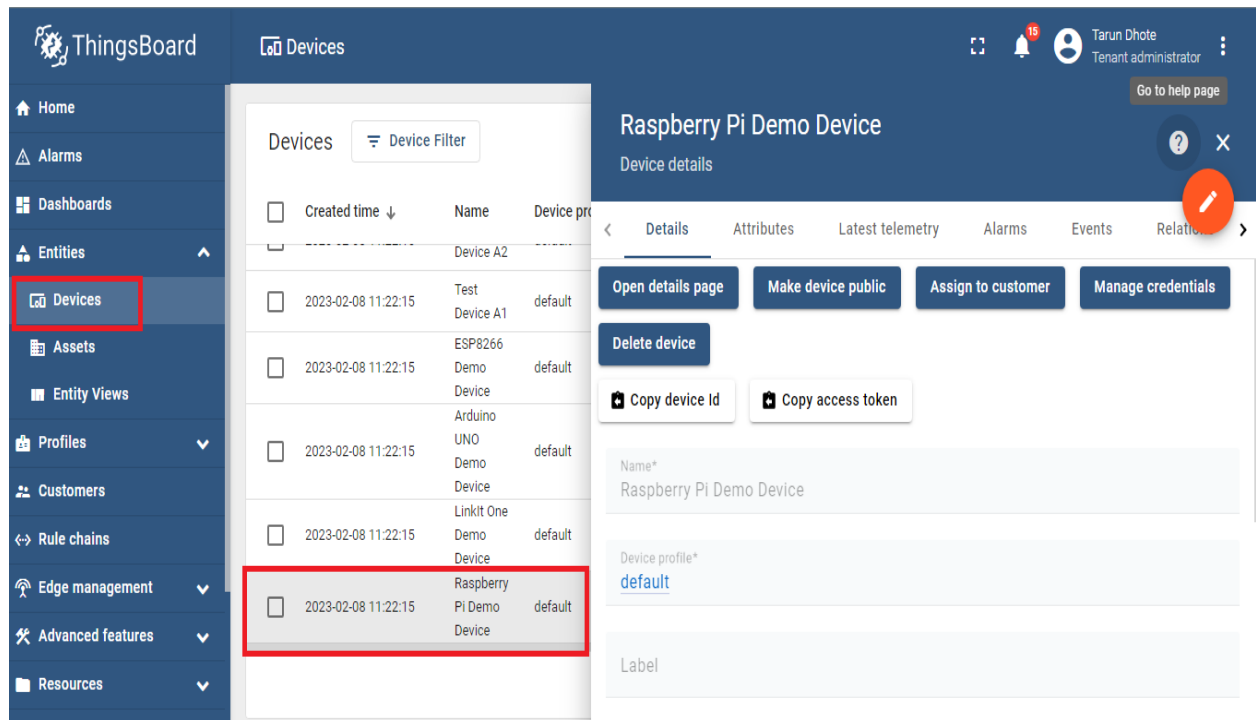


Step 2. Connect device

- Go to Devices → Demo Device

To connect the device you need to get the device credentials first. ThingsBoard supports various device credentials. We recommend using default auto-generated credentials which is access token for this guide.

- Click on the device row in the table to open device details. Note that the device state is "Inactive";
- Click "Copy access token". Token will be copied to your clipboard. Save it to a safe place.

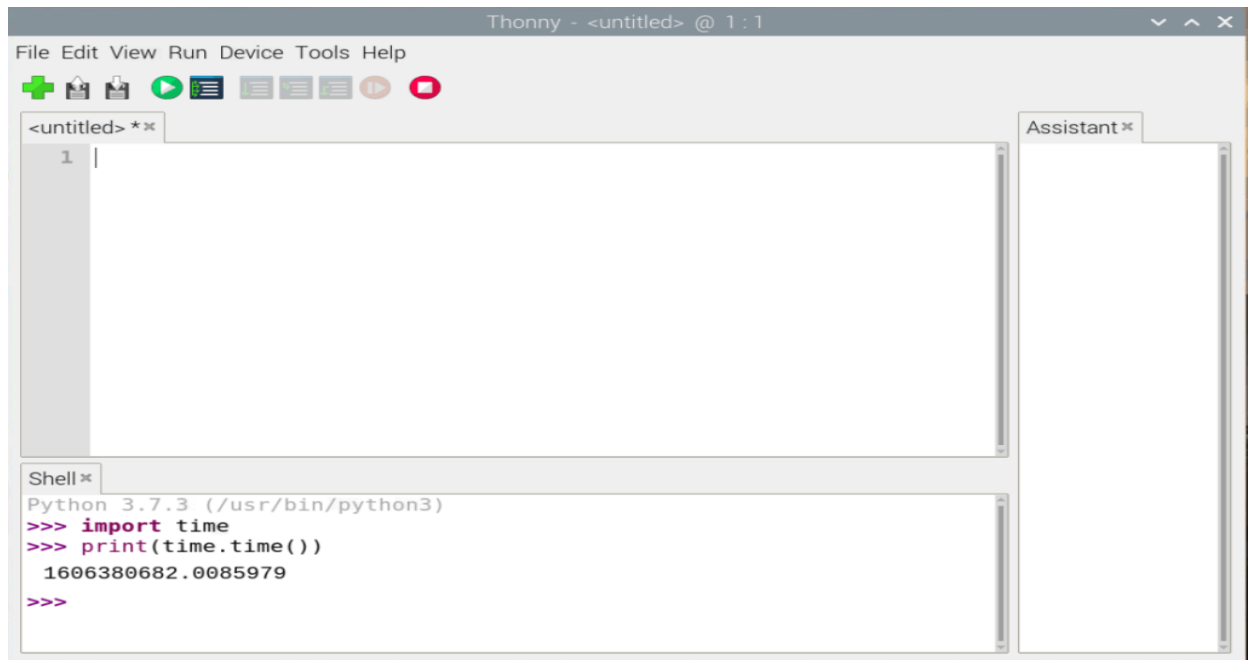


Programming the

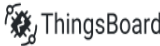
MQTT library installation

The following command will install MQTT Python library:

- **sudo pip install paho-mqtt**
- **Goto → home/logsun/Documents/Things-board**
- **Right click on rgb2.py code and open in Thonny IDE**



- Replace your access token with your copied access token previously you have done.



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[Architecture](#)
[API](#)

[mqtt-dht22.py](#)

```

1 import os
2 import time
3 import sys
4 import Adafruit_DHT as dht
5 import paho.mqtt.client as mqtt
6 import json
7
8 THINGSBOARD_HOST = 'demo.thingsboard.io'
9 ACCESS_TOKEN = 'DHT22_DEMO_TOKEN'
10 Replace your Access Token you have previously copied
11 # Data capture and upload interval in seconds. Less interval will eventu
12 INTERVAL=2
13
14 sensor_data = {'temperature': 0, 'humidity': 0}
  
```

- Provision your dashboard
- Programming the Raspberry Pi
- MQTT library installation
- Adafruit DHT library installation
- [Application source code](#)
- Running the application
- Data visualization
- See also
- Your feedback
- Next steps

- Click on Run to run the program

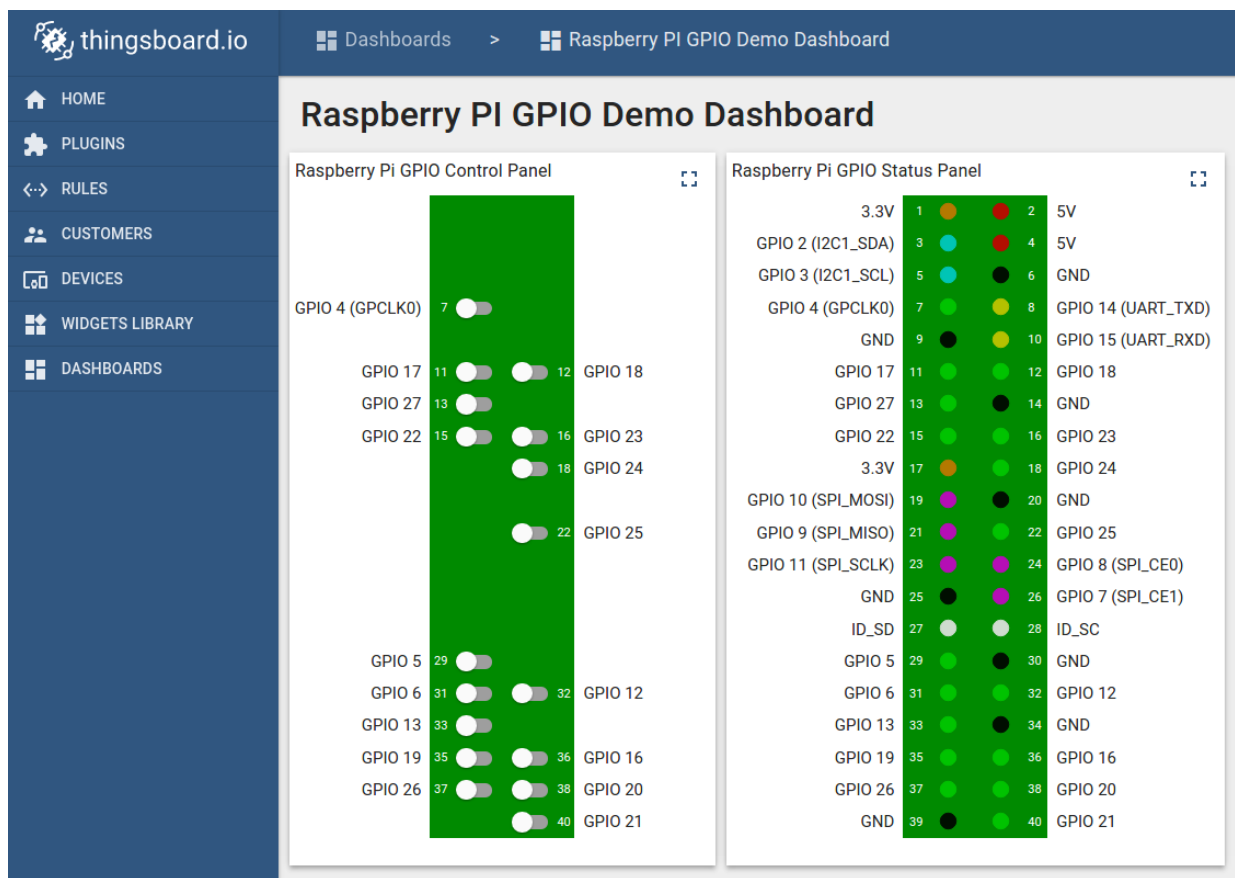
Data visualization

In order to simplify this guide, we have included “ GPIO Demo Dashboard” to the demo data that is available in each ThingsBoard installation.

in case of local ThingsBoard installation.

Once logged in, open **Dashboards-> GPIO Demo Dashboard** page. You should observe demo dashboard with GPIO control and status panel for your device. Now you can switch status of GPIOs using control panel. As a result, you will see LEDs status change on the device and on the status panel.

Below is the screenshot of the “ GPIO Demo Dashboard”.



4. Set up your XBee devices

Set up your XBee devices

1. Before you begin
2. Assemble the hardware
3. Download and install XCTU
4. Set up your first wireless connection

Step1: Download and install XCTU

This section contains download and install instructions based on operating system. XCTU is compatible with Linux, OSX, and Windows. It may be necessary to configure your system prior to installing XCTU for the first time.

- Install XCTU - Windows

Step2: Before you begin

- **Connect board with your laptop or PC.**
- **After that connect the XBee module to the board, follow the wired connections below.**

R-Pi	XBee
GND	GND
TX	TX
RX	RX
5V	VCC

Step3: Set up your first wireless connection

This section shows you how to configure two XBee modules in AT (transparent) mode. The XBee module passes information along exactly as it receives it. All serial data received by the XBee module is sent wirelessly to a remote destination XBee module.

- **Add XBee modules to XCTU**

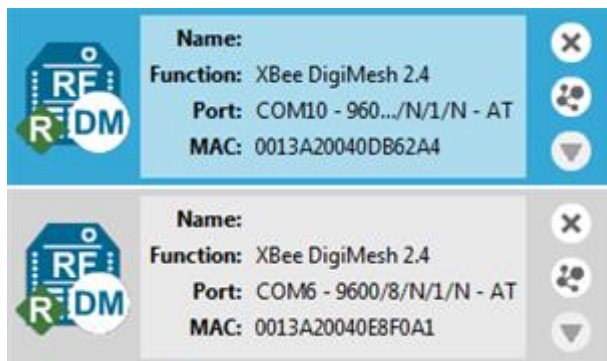
These instructions show you how to add two XBee modules to XCTU. However, you can use these instructions to add any number of XBee modules.

1. Connect two XBee modules to your computer using RASPBERRY PI boards.

2. Launch XCTU.
3. Click the **Configuration working modes** button .
4. Click the **Discover radio modules** button .
5. In the **Discover radio devices** dialog, select the serial ports where you want to look for XBee modules, and click **Next**.
6. In the **Set port parameters** window, maintain the default values and click **Finish**.

As XCTU locates radio modules, they appear in the Discovering radio modules dialog box.

7. Click **Add selected devices** once the discovery process has finished.
- You should see something similar to the following example in the **Radio Modules** section:



8. Click **Finish**.

Optional: Add a radio module

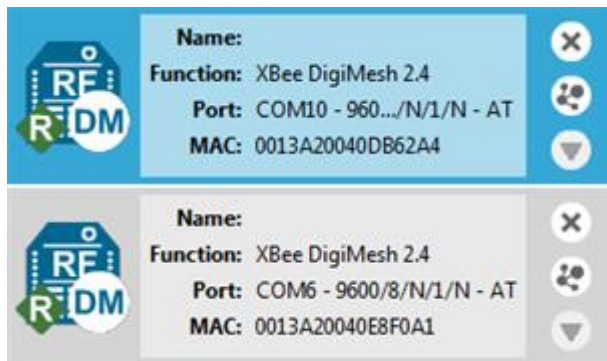
1. Click the **Add a radio module specifying the port settings** button .
2. In the **Add radio device dialog**, select the COM port for the device you want to add.
3. Click **Finish**.

- **Configure the first two XBee modules in transparent mode**

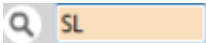
To transmit data wirelessly between your XBee modules, configure them to be in the same network.

Set up the first XBee module (XBEE_A):

Select the first XBee module.



Click the **Load default firmware settings** button .

Tip In the following steps, type parameter initials in the Search box to quickly find a parameter, as shown in the following example: 

Configure the following parameters for Co-Ordinator:

- **PAN id** = **1234**
- **JV** = **Disable**
- **Co-Ordinator CE** = **Enable**
- **DH** = **0**
- **DL** = **FFFF**
- **Node Name NI** = **Co-Ordinator**
- **AP** = **AT (Transparent Mode)**

Click the **Write radio settings** button .

Set up the second XBee module (XBEE_B):

Select the second XBee module.

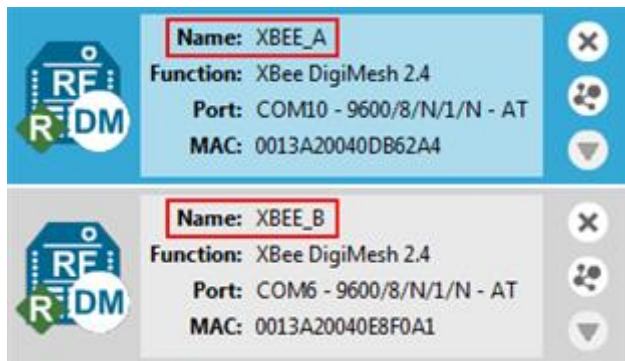
Click the **Load default firmware settings** button .

Configure the following parameters:

- **PAN id** = **1234**
- **JV** = **Enable**
- **Co-Ordinator CE** = **Disable**
- **DH** = **0**
- **DL** = **FFFF**
- **Node Name NI** = **Router**
- **AP** = **AT (Transparent Mode)**

Click the **Write radio settings** button .

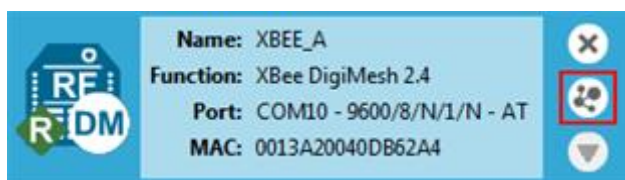
After you write the radio settings for the XBee modules, their names appear in the Radio Modules area.



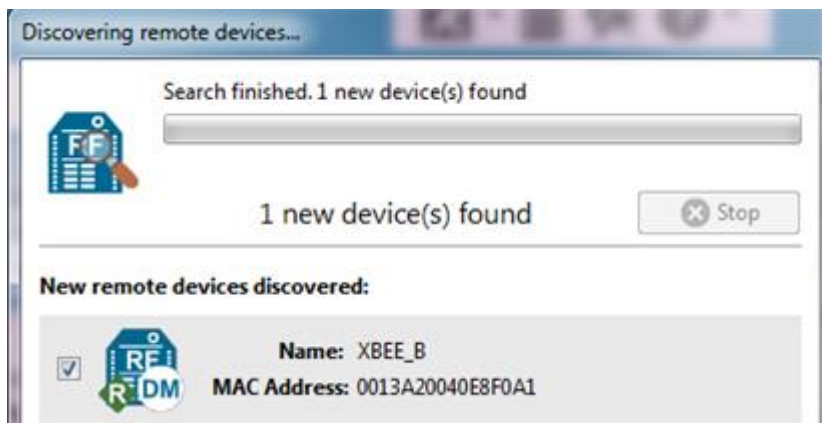
- **Check the network**

Once both XBee modules are configured, use XCTU to check that they are in the same network and can see each other.

Click the **Discover radio nodes in the same network** button  of XBEE_A.



The device searches for devices in the same network.



When the discovery process is finished, XCTU lists discovered devices found within the network in the Discovering remote devices dialog.

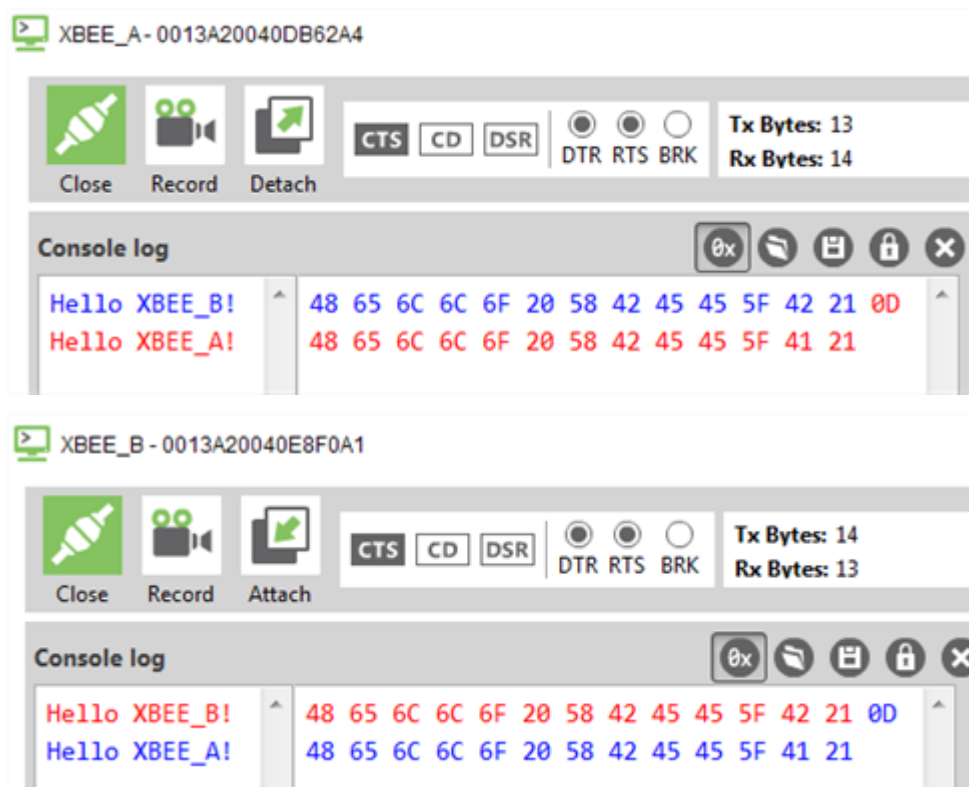
Click **Cancel**. There is no need to add the remote device that has been discovered.

- **Send messages through XCTU**

Use the XCTU console to have your two XBee modules send messages to each other.

1. Switch both XBee modules to the consoles working mode .
2. Open a serial connection for each XBee.
 - a. Select **XBEE_A** and click .
 - b. Select **XBEE_B** and click .
3. Click the **Detach view** button  to see both consoles at the same time.
 - a. In the **Console log** area for XBEE_A, type "Hello XBEE_B!"
 - b. In the **Console log** area for XBEE_B, type "Hello XBEE_A!"

The message of the sender is in blue font, and the message of the receiver is in red font.



4. Close the window for XBEE_B.
5. Keep the serial connections open  for both XBee modules.

Tip If the two XBee modules are unable to talk to each other:

- Verify that you accurately configured the parameters. See Configure the first two XBee modules in transparent mode.
- Verify that the **CH** (Operating Channel) and **ID** (Network Identifier) are the same for both XBee modules.

5.EXPERIMENTS

EXPERIMENT No 1- LED LOGIC & SWITCH

AIM:- To study the interfacing of LED and SWITCH to the .

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 9 pin female to female connector, 10 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi.
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And , 9 pin female to female connector, 10 pin female to female connector from the board J8 & J10 connector to “LEDS-SWITCHES” section.
- 5) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**.
- 8) To run this specific program enter the command >> **python3 ledSwitchLogic.py**
- 9) If you want to see program enter the command >> **sudo nano ledSwitchLogic.py**
- 10) To exit from the program press **Ctrl-Z**.
- 11) You will observe that LEDs will glow when you **ON** the switches and OFF when you **OFF** the switches.

Note: FRC port connection must be proper to run the program.

EXPERIMENT No 2.

INTERFACING LCD DISPLAY WITH RASPBERRY PI

AIM: - To study the interfacing of LCD (16*2) with the .(4 bit)

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 8 pin female to female connector, 5v,2A power supply, LCD.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect 26 pin FRC cable from the board J11 connector to “LCD-KEYPAD” section.
- 5) Open the Terminal window of and enter the command >> **cd Documents/IOT_SHALA**
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >> **python3 lcd.py**
- 9) If you want to see program enter the command >> **sudo nano lcd.py**
- 10) To exit from the program press **Ctrl-Z**.
- 11) You will observe that it displays “**LOGSUN SYSTEMS**” on LCD display.

THEORY:

Liquid crystal display:

LCD can be connected to the through the GPIO pins. LCD is connected in the 4-bit mode or 8-bit mode. And the standard subroutine is given with the development board. So that the application can be easily demonstrated and also for further implementation the subroutine can be easily embedded for which one has to do very few changes. Wide range of instruction functions: Clear displays, cursor home, display ON/OFF, cursor ON/OFF, cursor shift, display shift.

Experiment No 3.

INTERFACING DC MOTOR TO RASPBERRY PI

AIM: To study the interfacing of dc motor to Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply, DC motor, external12v dc power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect 9 pin female to female connector from the board J4 connector to “DC MOTOR” section.
- 5) Open the Terminal window of and enter the command >> **cd Documents/IOT_SHALA.**
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >> **python3 dcmotor.py**
- 9) If you want to see program enter the command >> **sudo nano dcmotor.py**
- 10) To exit from the program press **Ctrl-Z**.
- 11) You will observe that motor will rotating in clockwise direction for 5 seconds and Anticlockwise for 5 seconds

Experiment No 4.

INTERFACING RELAY WITH RASPBERRY PI

AIM: To study interfacing of relay with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 5 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect 5 pin female to female connector from the board J12 connector to "RELAY" section.
- 5) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA.
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >>
python3 relayControlFromCloud.py
- 9) If you want to see program enter the command >>
sudo nano relayControlFromCloud.py
- 10) To exit from the program press **Ctrl-Z**.
- 11) You will observe that relay is ON for 2 sec and OFF for 2 sec.

Experiment No 5.

INTERFACING BUZZER WITH RASPBERRY PI

AIM: To study interfacing of relay with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 5 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect 5 pin female to female connector from the board J12 connector to "BUZZER" section.
- 5) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA.
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >>
python3 buzzerControlFromCloud.py
- 9) If you want to see program enter the command >> **sudo nano relay.py**
- 10) To exit from the program press **Ctrl-Z**.
- 11) You will observe that buzzer is ON for 2 sec and OFF for 2 sec.

Experiment No 6.

INTERFACING ULTRASONIC SENSOR WITH RASPBERRY PI

AIM: To study interfacing of ultrasonic sensor with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect 4 pin female to female connector from the board J7 connector to "ULTRASONIC SENSOR" section.
- 5) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA.
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >>
python3 ultrasonicSendToCloud.py
- 9) If you want to see program enter the command >>
sudo nano ultrasonicSendToCloud.py
- 10) To exit from the program press **Ctrl-Z**.
- 11) Open thingsboard → Goto Devices → select your created device → latest telemetry
- 12) You will observe that ultrasonic distance readings upload on thingsboard dashboard.

Experiment No 7.

INTERFACING PIR SENSOR WITH RASPBERRY PI

AIM: To study interfacing of PIR sensor with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect pin female to female connector from the board J3 connector to "PIR SENSOR" section.
- 5) Open the Terminal window of and enter the command >> **cd Documents/IOT_SHALA.**
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >> **python3 pirSendToCloud.py**
- 9) If you want to see program enter the command >> **sudo nano pirSendToCloud.py**
- 10) To exit from the program press **Ctrl-Z**.
- 11) Open thingsboard → Goto Devices → select your created device → latest telemetry
- 12) You will observe that PIR data upload on thingsboard dashboard.

Experiment No 8.

INTERFACING RGB SENSOR WITH RASPBERRY PI

AIM: To study interfacing of RGB sensor with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi.
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect pin female to female connector from the board J6 connector to “RGB SENSOR” section.
- 5) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA.
- 6) Now we are into the folder **IOT_SHALA**
- 7) To see the list of files into **IOT_SHALA** folder enter command >> **ls**
- 8) To run this specific program enter the command >>
python3 rgbControlFromCloud.py
- 9) If you want to see program enter the command >>
sudo nano rgbControlFromCloud.py
- 10) To exit from the program press **Ctrl-Z**.
- 11) Open thingsboard → Goto Dashboard → GPIO Dashboard(Deprecated)
- 12) You will observe that **GPIO Control Panel** and **GPIO Control Panel** will seen on screen
- 13) Click on GPIO to on and off the GPIO's and to change the state of RGB.

Experiment No 9.

INTERFACING DHT22 SENSOR WITH RASPBERRY PI

AIM: To study interfacing of PIR sensor with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi.
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect pin female to female connector from the board J4 connector to "DHT11/22" section.

Run commands through terminals

- 5) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA

Now we are into the folder / **IOT_SHALA**

To see the list of files into / **IOT_SHALA** folder enter command >> **ls**

To run this specific program enter the command >>

python3 dhtSendToCloud.py

- 6) To exit from the program press **Ctrl-Z**.
- 7) Open thingsboard → Goto Devices → select your created device → latest telemetry
- 8) You will observe that DHT11/22 data upload on thingsboard dashboard.

Experiment No 10.

INTERFACING XBee WITH RASPBERRY PI

AIM: To study interfacing of PIR sensor with Raspberry pi

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi.
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect pin female to female connector from the board J7 connector to "XBEE Module".

R-Pi(J7)	XBee
GND	GND
TX-15	TX
RX-14	RX
5V	VCC

- 5) open terminal window
- 6) type command
sudo raspi-config
- 7) Select "Interface Options"
- 8) Select "I6 Serial Port"
- 9) A new window will and ask for " Would you like a login shell to be accessible over serial?"
Select <No>

10) A new window will ask for “Would you like the serial port hardware to be Enabled”

Select <Yes>

Select <Ok>

11) Run commands for both transmitter and receiver through their respective terminals

12) For Zigbee Receiver Open the Terminal window of and enter the command >>

cd Documents/IOT_SHALA/XBee/XBee_Rx

Now we are into the folder **/XBee_Rx**

To see the list of files into **/XBee_Rx** folder enter command >> **ls**

To run this specific program enter the command >> **python3 XBee_Recv.py**

13) For Zigbee Transmitter Open the Terminal window of and enter the c command >>

cd Documents/IOT_SHALA/XBee/XBee_Tx

Now we are into the folder **/XBee_Tx**

To see the list of files into **/XBee_Tx** folder enter command >> **ls**

To run this specific program enter the command >> **python3 XBee_trans.py**

14) To exit from the program press **Ctrl-Z**.

Experiment No 11.

INTERFACING BLUETOOTH SENSOR WITH RASPBERRY PI

AIM: To study interfacing of Bluetooth with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi .
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect pin female to female connector from the board J6 connector to “RGB” section.
- 5) Need to install bluedot libraries, BlueDot allows you to control your projects wirelessly
 - Open the Terminal window of and Install the Python library:

sudo pip3 install bluedot

To upgrade to the latest version:

sudo pip3 install bluedot --upgrade

- Get the Android Blue Dot app or use the Python Blue Dot app on your Android Phone.

Note : You don't need to install the library all the time, once library is installed you only need to pair your android phone to with Bluetooth.

- 6) Pair a and Android phone

On your Android phone:

1. Open Settings
2. Select Bluetooth and make your phone “discoverable”

On your :

1. Click Bluetooth ▸ Turn On Bluetooth (if it's off)

2. Click Bluetooth ▸ Make Discoverable
3. Click Bluetooth ▸ Add Device
4. Your phone will appear in the list, select it and click **Pair**

On your Android phone and .

1. Confirm the pairing code matches
2. Click OK

7) Using the Command Line

On your Android phone:

1. Open Settings
2. Select Bluetooth and make your phone “discoverable”

On your :

1. Open terminal and type “**sudo bluetoothctl**” and press Enter to open Bluetooth control
2. At the [bluetooth]# prompt enter the following commands:

discoverable on

pairable on

agent on

default-agent

scan on

3. Wait for a message to appear showing the Android phone has been found:
4. [NEW] Device 12:23:34:45:56:67 (MAC ADDRESS) devicename
5. Type pair with the mac address of your Android phone:
6. pair (MAC ADDRESS)

On your Android phone and .

1. Confirm the passcode.
2. Type **quit** and press Enter to return to the command line

8) Connecting

- 9) Run commands through terminals

- 10) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA

Now we are into the folder **/ IOT_SHALA**

To see the list of files into **/ IOT_SHALA** folder enter command **>> ls**

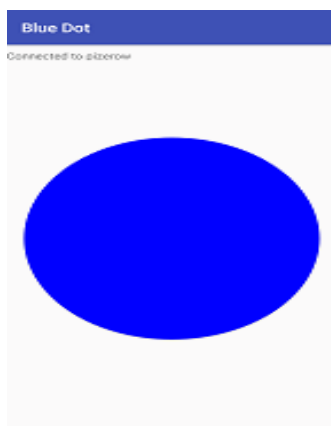
To run this specific program enter the command **>>**

python3 Test_Bluetooth_Button.py

- 11) To exit from the program press **Ctrl-Z**.
- 12) Start-up the Blue Dot app on your Android phone
 - Select your from the list



- Press the Blue Dot



Note

Your python program will need to be running and Waiting for connection before the BlueDot app will be able to connect to your

6. FRC Connector Details

A) 40 PIN FRC CONNECTOR DETAILS (J7) :

PIN NO.	DESCRIPTION	PIN NO.	DESCRIPTION
1	SCL	2	SDA
3	GP27	4	GP22
5	GP16	6	GP10
7	GP12	8	GP20
9	GP7	10	GP21
11	GP8	12	GP26
13	GP25	14	GP19
15	GP24	15	GP13
17	GP23	18	GP6
19	TX	20	GP5
21	RX	22	GP11
23	GP18	24	GP9
25	GND	26	5V

No	Experiments	Peripherals	RPi-Board
1	LED LOGIC SWITCH	SV8 SV10	J10 J8
2	LCD Display	SV4	J11
3	DC Motor	SV1	J4
4	Relay	SV13	J12
5	Buzzer	SV13	J12
6	Ultrasonic Sensor	SV6	J7
7	PIR sensor	SV12	J3
8	RGB	SV11	J6
9	DHT22	SV9	J4
10	XBee	XBee Module	J7